

2 Trees: Build Your (D)Home

One small chainsaw, 120 wedge struts, and a geodesic dome in a fortnight

Built 2026-09-08. Every figure in this book was computed at build time from the geometry in the DomeSim project.

Contents

Part 1 · Two Trees

- | | | |
|---|--------------------------------|---------|
| 1 | The Tree That Was Already Down | story |
| 2 | Fifteen Middlemen | explain |
| 3 | The Frankendome | story |
| 4 | The V-Bracket Afternoon | story |
| 5 | The Cut That Changed It | story |

Part 2 · Why a Dome Lets You Do This

- | | | |
|----|--|---------|
| 6 | Triangles Do Not Care What Shape Your Stick Is | explain |
| 7 | Not a Worse Two-by-Four | explain |
| 8 | Forty Panels, One Hundred and Twenty Members | explain |
| 9 | Two Lengths, Not Forty | explain |
| 10 | Compression, Tension, and One Eighth of a Tree | explain |
| 11 | What a Dome Costs You | explain |

Part 3 · The Wedge

- | | | |
|----|-------------------|-------|
| 12 | Split Like a Cake | howto |
|----|-------------------|-------|

13	Eighty-Seven Percent	explain
14	Four Ways Up	explain
15	The Pinwheel	explain
16	The Seam and Its Key	explain
17	Let the Connector Hold the Angle	explain
18	Why Every Panel Keeps Its Own Edge	explain
19	Bends, Knots and Small Units	explain

Part 4 · The Two Calculations

20	Which Way Round Are You Working?	howto
21	Method A: From the Dome You Want	howto
22	Method B: From the Tree You Have	howto
23	The Round Trip	explain
24	The Worked Build	reference
25	When the Numbers Say No	howto

Part 5 · The Fortnight

26	Day Zero: The Saw	story
27	Days One and Two: Felling	howto
28	Days Three and Four: Bucking	howto
29	Days Five to Eight: Ripping	howto
30	Day Nine: The Jig	howto
31	Days Ten to Twelve: Forty Panels	howto
32	Days Thirteen and Fourteen: Raising	howto

Part 6 · The Jig and the Cuts

33	One Bench, Forty Panels	reference
34	The Butt Cut	reference

35	The Head Overfit	reference
36	Flush-Cutting in Place	reference
37	Panel Drawings	reference
38	Where Error Goes	explain

Part 7 · From Frame to Home

39	The Footing and the Ring	howto
40	Closing the Shell	howto
41	Doors, Windows and the Rim	howto
42	Weather and Water	howto
43	Inside a Round Room	explain
44	Heat, Power and the Small Systems	howto

Part 8 · What It Actually Cost

45	The Money, Both Ways	reference
46	The Hours	reference
47	What Broke	story
48	Corrections	reference

Part 9 · Take It Further

49	Bigger, Smaller, Other Frequencies	explain
50	Other Species, Other Sections	howto
51	Building With Other People	howto
52	What I Would Do Differently	story

Title pages

Half-title

The title alone, on a field of bark.

PHOTOGRAPH

(no caption yet)

*Close bark texture, shallow depth of field. No dome
visible yet.*

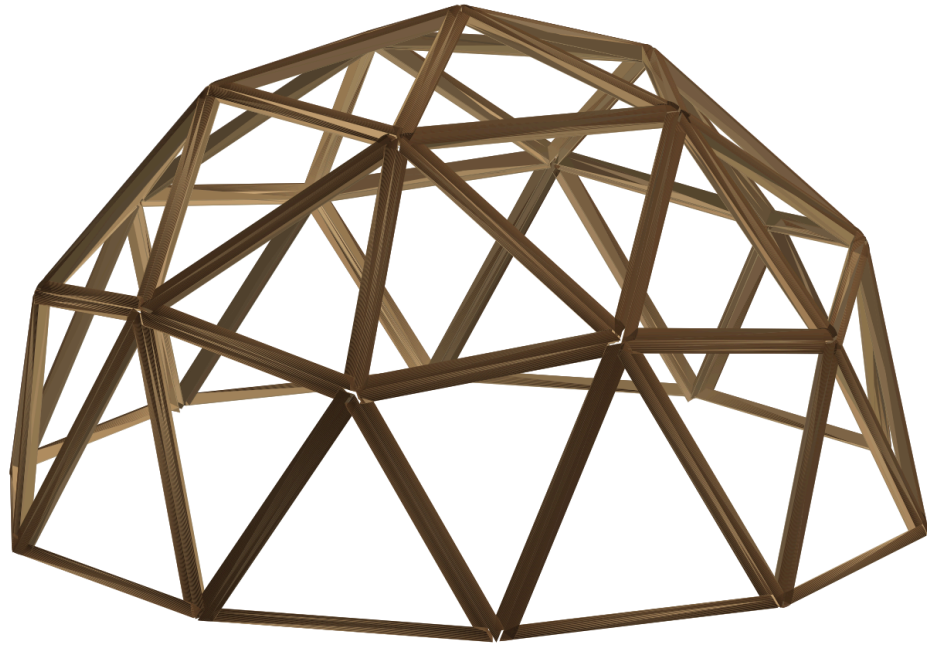
front-bark



(not written yet)

Frontispiece

The finished frame against sky, so the reader knows on page two what the book is for.



The frame, day fourteen.

(not written yet)

Title page

Title, subtitle, author, imprint.

(not written yet)

Copyright and colophon

Rights, edition, and the fact that every number in the book was generated by the software listed in the back.

(not written yet)

What this book is

Two pine trees, one chainsaw off a hardware-store shelf, one hundred and twenty structural members, and a geodesic dome standing in fourteen days.

That is the whole claim, and everything in this book exists to either explain it, prove it, or teach you to do it.

It is a real build

The dome in the photographs was not rendered, proposed, or costed on a spreadsheet. It was cut, and the process of cutting it produced a number of surprises, most of which are in here. Several chapters exist specifically to report a figure that does not flatter the method — Chapter 13 revises the recovery claim downward, Chapter 45 halves an hourly rate this project publicised, and Chapter 48 lists everything else that has had to be taken back. A book about a new building method that contains no corrections is a brochure.

It is three books at once

The reader I had in mind wanted three different things, and they read at three different speeds, so rather than blend them into a mush each chapter declares which one it is.

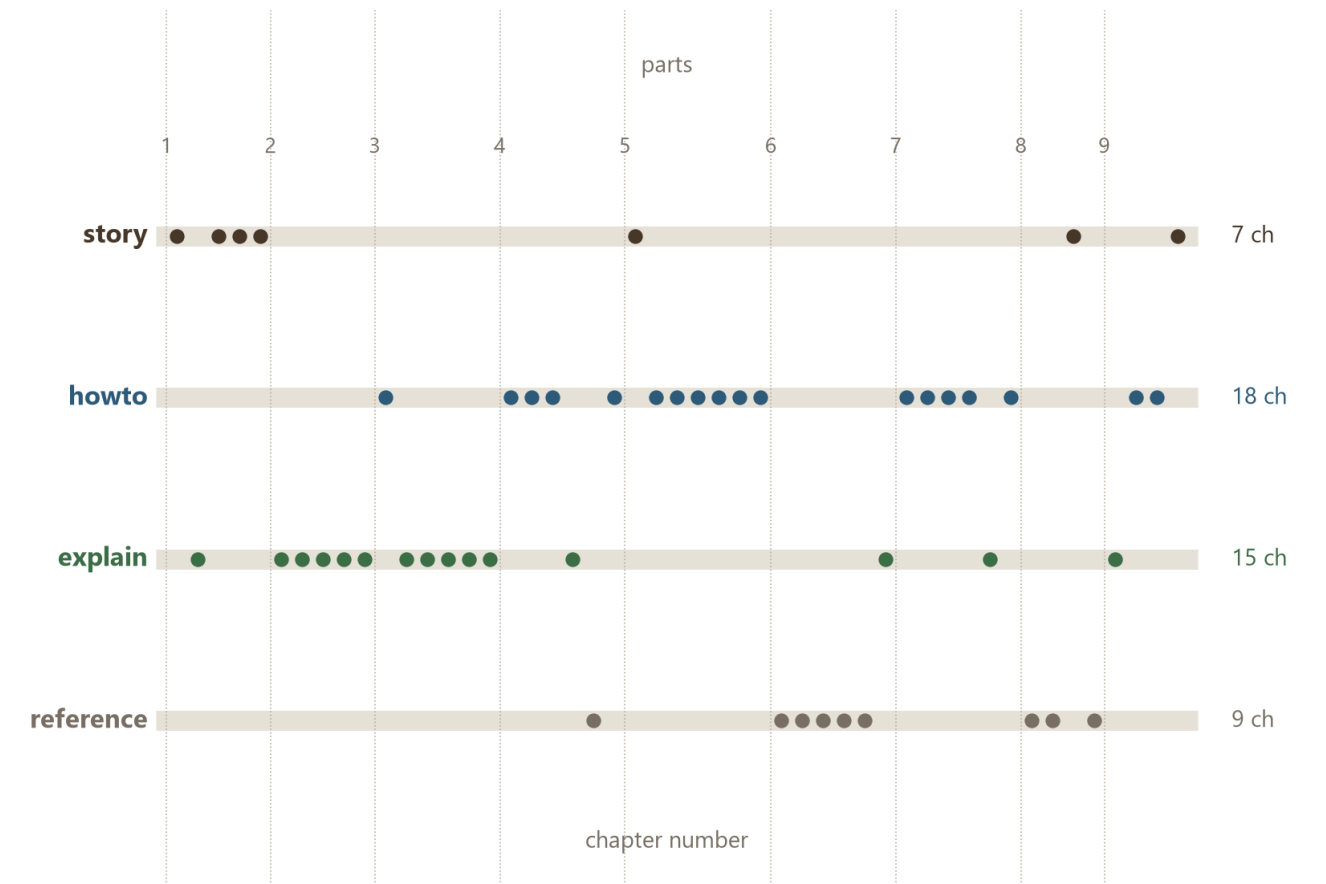
The story is what happened, in order, to somebody with a saw: the frankendome that came first, the afternoon the wedge was found, and the week lost to a problem that turned out not to exist. Seven chapters, and you can read them straight through as a narrative.

The manual is do this, then this. Eighteen chapters, and if you follow only those you get a dome. It is a complete set of instructions on its own.

The explanation is why any of it works — why a triangulated frame will accept a stick no mill would sell, why splitting a round log is free and squaring it is not, and why this method does not transfer to a rectangular house. Fifteen chapters, all skippable, all there because a method you do not understand is a method you cannot adapt.

The rest is reference: the tables you come back to at the saw rather than read in an armchair.

Three ways through this book



Every chapter belongs to one strand. Read straight down the book, or follow one track. The manual strand alone is a complete set of instructions.

Three ways through this book.

Every number here is computed

This is the unusual part, and it is worth a paragraph.

Not one figure in this book was typed into a sentence. Every dimension, count, angle, percentage and price is a token in the manuscript that gets filled in, at the moment the book is made, by the same software that draws the three-dimensional models. When it says the dome is 21.6 feet across, that number came out of a solver about a second before it reached this page.

The reason is not showing off. It is that a number typed into a paragraph is correct on the day it is typed and silently wrong from the first time anybody changes anything — and there is no way to find those. Nothing fails. The book just quietly becomes untrue in places nobody remembers.

So the arithmetic in this book is a test suite. It runs, it checks itself against the geometry, and where two ways of computing something disagree the book prints both and says which is which. Chapter 23 is nothing but a proof that the two methods in Part IV are the same calculation, and it prints the residual — 0.0e+00 inches — rather than asking you to take it on trust.

Every tool that produced these numbers is named in the back, and every one of them is included with this book. You can change the tree and rebuild the whole thing around your own.

What it is not

It is not an engineer's stamp. Nothing here has been checked by a licensed structural engineer, and where a figure depends on species, grade or moisture this book says so and tells you who to ask instead of guessing.

It is not a code compliance document. Your jurisdiction has opinions about round buildings and they are not in here.

And it is not a promise that you will do this in a fortnight. I did, on the second attempt, having already made every mistake in Chapter 47.

PART 1

Two Trees

“I cut a tree, I rip the trunk, it is usable wood. Fifteen middlemen, gone.”

The story strand. How the method was found, and what it replaced.

CHAPTER 1 · STORY

The Tree That Was Already Down

A windfall pine, a small saw, and a question nobody had asked out loud

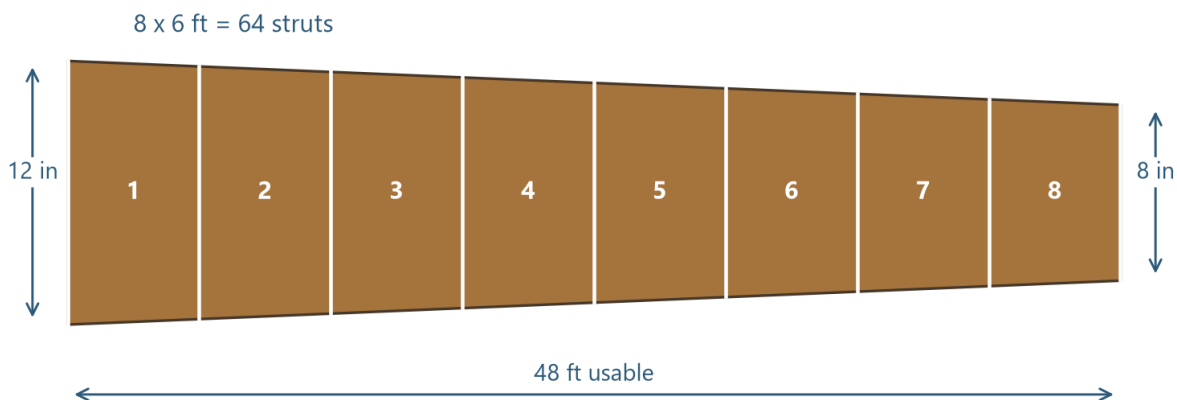
This chapter is planned but not written yet. Its page plan is in the outline.

The Tree That Was Already Down

Sixty feet of pine, and no way to use it

The tree, measured

The book's tree, measured



48 feet of usable trunk, 12 inches at the butt and 8 at the top, bucked into 8 sections of 6 feet. Every number in this book descends from these three measurements.

The book's tree: 12 inches at the butt, 8 at the top, 48 feet of usable trunk.

What I actually wanted

Fifteen Middlemen

Why a two-by-four costs what it costs, and how much of that is wood

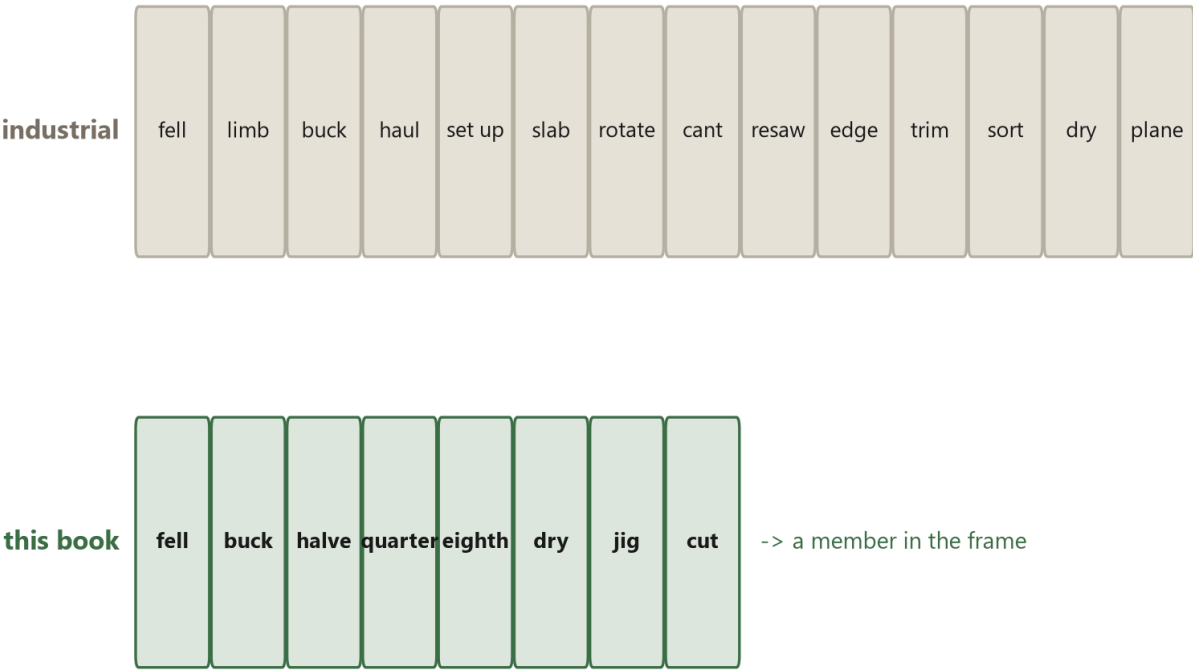
This chapter is planned but not written yet. Its page plan is in the outline.

Fifteen Middlemen

From stump to rack

The chain, and where the wood goes

Two routes from a standing tree to a structural member



14 operations against 8. The wedge route does not do any of it better -- it skips it, and it can only skip it because the dome will accept the shape that falls out of a split log. The two lists do not stop in the same place, and that is stated rather than hidden: the mill route ends at a graded board, which still has to be bought, carried and jointed. The wedge route already includes its joint cuts.

Every step between a standing tree and a board on a rack.

The price of a 2x4, taken apart

Nominal, dressed, and the missing third

The Frankendome

Forty triangles, no two struts alike, and the failure that taught the method

This chapter is planned but not written yet. Its page plan is in the outline.

The Frankendome

Strut-agnostic, and what that cost

The frankendome standing

Figure `franken-standing` has not been rendered yet.
The frankendome: it stood, and no two members matched.

The ten-piece pentagon

Why I stopped

CHAPTER 4 · STORY

The V-Bracket Afternoon

Trying to join shapes that have nothing in common

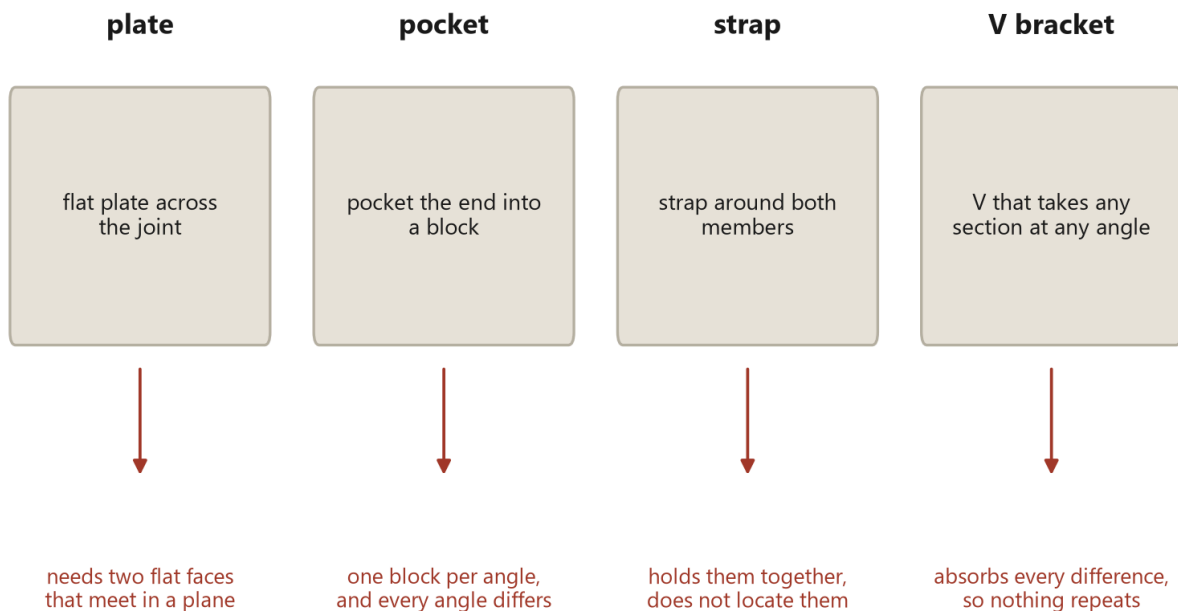
This chapter is planned but not written yet. Its page plan is in the outline.

The V-Bracket Afternoon

A bracket that does not care

Four joints that nearly worked

Four joints that nearly worked



The V bracket was the best of them, and that was the problem: a connector that tolerates any difference removes the pressure to make the parts the same. The wedge went the other way -- make the parts identical and the joint gets to be simple.

Four joints, and the one thing each could not do.

The question that ended it

The Cut That Changed It

Split the trunk like a cake, and the stick is already the right shape

The Cut That Changed It

I want to tell this one without hindsight tidying it up, because the tidy version — *I realised the dome would accept any consistent section, therefore I split the log radially* — is not what happened and it teaches nothing. What happened was that I was tired of brackets.

Half, half, half again

The frankendome had taught me that a frame of forty triangles will stand up even if no two of its sticks match. The V-bracket afternoon had taught me that making a connector tolerant enough to join anything to anything means the connector is now doing all the work, forever, in forty places.

So I had stopped asking how to join dissimilar sticks and started asking the opposite question: what would it take for them not to be dissimilar?

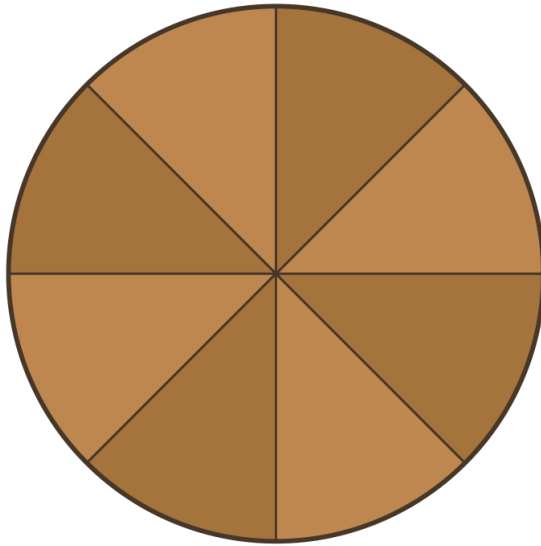
The answer that everybody gives is *mill them*. Square them up, run them through a planer, and now they are all 2x4s. That answer costs a mill, a planer, a shed to keep them in, and about half the tree.

I was sitting on a section of trunk at the time. Not thinking about it. Sitting on it.

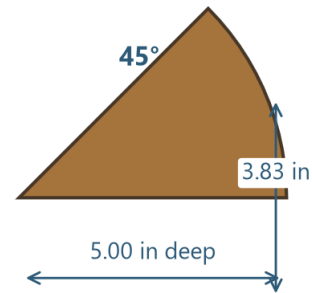
And the thing about a round section of trunk is that it is already rotationally symmetric. Every radius is the same as every other radius. If you split it through the middle you get two identical halves, not because you were careful but because the log was round before you touched it. Split those and you get four. Split those and you get 8.

Three passes. Eight sticks. Every one the same, to whatever tolerance the tree was round to, which is a tolerance nobody had to hold and nobody had to check.

One trunk, eight sticks



8 sectors, 3 saw passes



one member, as it comes off the saw

A 10-inch section split 8 ways gives 8 members of 3.83 by 5.00 inches, 9.8 square inches each. Nothing is squared, edged or planed. The bark face stays curved.

Three cuts. Eight structural members.

I did it that afternoon on a piece I had been about to burn.

What became obvious immediately

Four things, in about a minute, and I want to record that they were obvious because the ones in the next section took a week and I do not want to flatter the story.

One section, everywhere. Every stick in the frame has the same cross-section: 3.83 inches across the bark face, 5.00 inches from pith to bark, 9.82 square inches of wood. So one jig fits all of them. One fixture, one set of stops, one procedure repeated a hundred and twenty times. That is what makes this a fortnight instead of a summer.

Depth where you want it. A sector is deep — it runs the whole way from the pith to the bark — and depth is where bending stiffness lives. The stick is not a compromise between what the tree gave and

what a mill wanted. It is the deepest member you can cut from that log, which is a strange thing to get for free.

No machine between the tree and the frame. Fell, buck, split, build. The chainsaw does all of it. There is no second machine, no second setup, no second building to keep the second machine in.

Four ways up. This one I noticed while stacking them. The sector is not symmetric about its own long axis — it has a point and a curved back — so you can rotate it four sensible ways in the wall and get four different things: which face the weather sees, which face the next stick lands on, whether the points of a pair aim together or apart. That is Chapter 14, and it is the most genuinely new thing in the method.

And what took another week

Now the part the tidy version leaves out.

I assumed, because the sticks were identical and the panels were identical, that the *joints* would be identical. Forty triangles, all the same, all meeting each other the same way.

They do not.

A dome is a sphere approximated by flat panels, and the angle at which two panels meet is not the same everywhere on it. It cannot be — the sphere curves differently in different places, and a 2V hemisphere has two classes of edge for exactly that reason. So the seam between two panels folds by one amount here and a different amount there, and a key cut to fit one seam will not fit the other.

I did not want this to be true. I spent the better part of a week trying to make it not be true: fudging the panel outlines, trying a compressible gasket, trying to talk myself into the idea that the difference was small enough to force closed. It is not small enough. Forty panels of *nearly* closing is a dome with a gap in it.

What actually resolved it was measuring instead of arguing. Across all 55 seams there are only **2** fold angles. Not forty. Not a continuum. Two: 18.03 degrees and 22.46.

That was the moment the method became buildable rather than merely elegant. 2 angles means 2 key profiles: cut one of each until it fits, then repeat it 25 and 30 times. The difference between “the seams vary” and “the seams take 2 values” is the difference between a research project and a fortnight, and I lost a week to not knowing which one I was in.

The chart is in Chapter 16, where it belongs.

The bit I got wrong in public

While all this was going on I was saying, out loud and on camera, that nothing in the dome needed a mitre.

That is not true and it was never true. The end of each stick that lands on the side of the next one — the butt end — is a compound cut: a mitre and a bevel together. What is true, and what I should have been saying, is that no end is ever mitred *to another end*. Every joint is an end landing on a flat side, which means each stick is cut once, on one end, before it goes anywhere near the assembly, and the other end is left long and sawn off in place.

That is a much better claim than the one I made, and it is a shame I spent several months making the wrong one. Chapter 15 has the geometry and Chapter 48 has the full list of things this project has had to take back.

What it is worth

The number that convinced me it was not just a neat trick: splitting this way keeps 87.4 per cent of the solid wood in the trunk, against about 45 per cent for the same log sawn into two-by-fours the way a mill would. That is 1.93 times the usable wood, from a cheaper process, with one machine instead of four.

Chapter 13 does that arithmetic properly, including the parts of it that are less flattering than they first appeared.

But sitting on that log, what I had was a much simpler observation, and I still think it is the whole book:

A round thing splits into identical pieces for free. A square thing has to be made square, and you pay for that in wood, in machines, and in time. Domes are the buildings that will accept the free pieces.

PART 2

Why a Dome Lets You Do This

“We can only use this method because of the dome's inherent structure.”

The explanation strand. Skip it if you only want the manual, but the method does not transfer to a square house and this is the chapter that says why.

Triangles Do Not Care What Shape Your Stick Is

Why a triangulated frame can accept a member no mill would sell

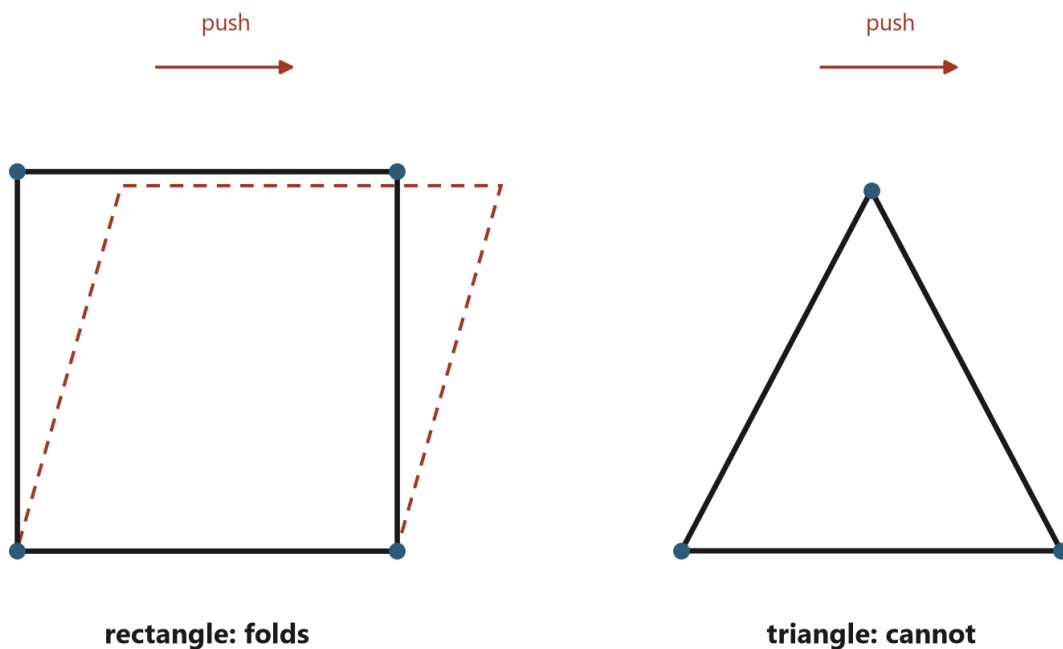
This chapter is planned but not written yet. Its page plan is in the outline.

Triangles Do Not Care

Axial, not bending

The same load, two frames

The same load, two frames



A rectangle of four pinned members is a mechanism: it folds. It needs a sheet, a brace or a stiff joint to stand. A triangle of three cannot change shape without one of them changing length, so the joint is only asked to hold the ends together -- which is why the shape of the stick stops mattering.

A rectangle needs a sheet to stand. A triangle does not.

What the wedge is good at

One eighth of a tree

Not a Worse Two-by-Four

The head-to-head nobody runs honestly, run honestly

Not a Worse Two-by-Four

Everybody asks the same question, and they are right to ask it.

Is a split log actually as good as a board?

The honest answer has three parts, and only two of them are flattering. This chapter gives all three, because a method defended only with its good numbers is a method nobody should trust with a roof over their head.

The rectangle is not the tree's idea

Modern construction has trained us to think of wood as something that naturally arrives in rectangles. A tree goes into a sawmill and a stack of 2x4s comes out, and the form is so familiar that it stops looking like a choice.

It is a choice. The rectangle is not a property of wood. It is the output of a manufacturing process designed around a particular kind of building — one made of vertical studs, horizontal plates, flat walls, right-angle corners and rectangular openings. For that building the rectangle is close to perfect, and the industry that produces it is very good indeed at what it does.

A geodesic dome is almost the opposite of that building. Its structural language is triangles, changing face planes, radial geometry and repeated angular joints. Nothing in it is a right angle. Nothing in it is a flat wall.

So the question reopens, and it is worth asking properly rather than assuming the answer in either direction: *for this building, is the rectangle still the right thing to make out of the tree?*

More wood, from a modest tree

Start with the simplest measure: how much wood is actually in the stick.

Take an 8-inch log — not a monster, a tree most people could fell and move without help. Split it 8 ways and each sector holds **6.28 square inches** of wood.

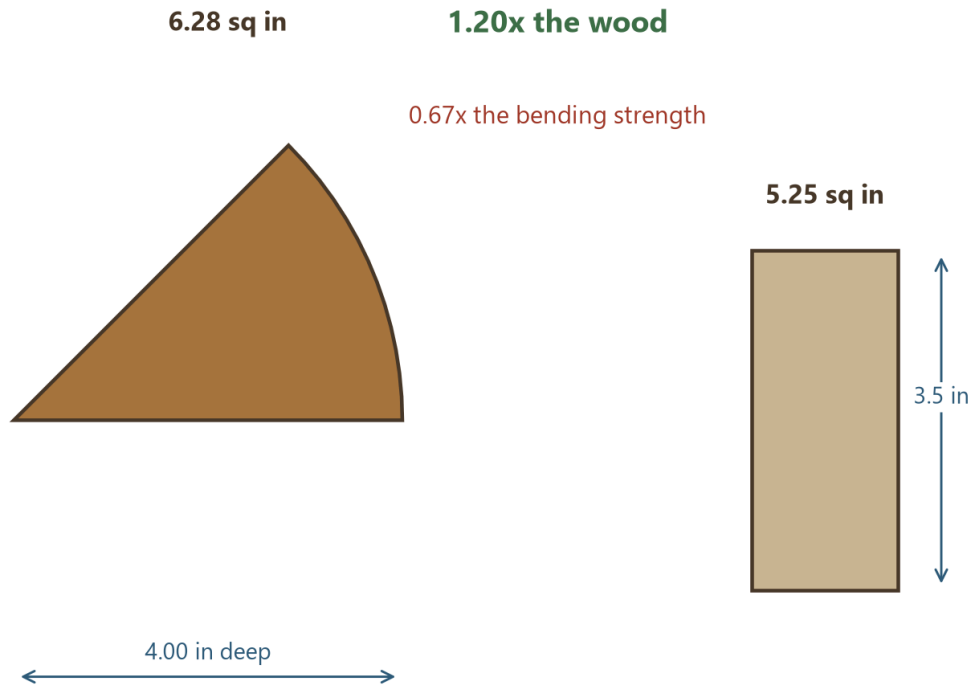
A dressed 2x4 — the board actually on the rack, which is 1.5 by 3.5 inches whatever it says on the label — holds **5.25**.

That is **+20 per cent more wood** in the wedge, from a tree you could carry out of a wood lot in sections.

And it is the conservative version of the claim. A larger log widens the gap sharply, because area grows with the square of the radius while a 2x4 stays a 2x4. The book quotes the small log deliberately: a claim is worth more made at its weakest.

The wedge is also deeper. Pith to bark is 4.00 inches against the board's 3.5, and depth is where bending resistance comes from.

The two sections, at the same scale



one eighth of the log

dressed 2x4, on edge

An eighth of an 8-inch log holds 6.28 square inches of wood against the dressed 2x4's 5.25 -- +20 per cent. It is also deeper: 4.00 inches pith to bark against 3.5. What it is not is stronger in bending, because so much of that wood sits near the pith, where it does the least work.

The two sections, at the same scale.

And now the number that does not help

Here is where an honest chapter parts company with a sales pitch.

Area is not strength. How much a section resists bending depends not only on how much wood is in it but on *where that wood sits* — material far from the middle does most of the work, material near the middle does almost none.

And a wedge puts a great deal of its wood near the pith, in the narrow part of the pie slice, where it contributes very little.

Run the numbers properly and the comparison comes out like this:

	wedge	2x4 on edge
Wood in the section	+20%	—
Resistance to bending (stiffness)	-1%	—
Bending strength	-33%	—

Read that middle row again. The wedge has +20 per cent more wood in it and is **not stiffer**. It comes out within a per cent of a 2x4 stood on edge — effectively identical, having spent twenty per cent more material to get there.

And the bottom row is worse: in bending *strength* the wedge is -33 per cent against the same board.

That is the number this method's advocates do not usually print, and it is printed here, in a table, near the front.

One eighth of an 8" log against a dressed 2x4

	wedge	2x4 on edge	ratio
Cross-section area	6.28 in ²	5.25 in ²	1.20x
Depth	4.00 in	3.50 in	1.14x
Width	3.06 in	1.50 in	2.04x
Second moment, strong	5.33 in ⁴	5.36 in ⁴	0.99x
Section modulus, strong	2.05 in ³	3.06 in ³	0.67x
Same board laid flat	5.33 in⁴	0.98 in⁴	5.41x

The wedge wins on area and loses on bending strength, and both are printed. Area is what an axial member trades on; section modulus is what a beam trades on. In a triangulated frame the members are mostly axial, which is the argument -- but panels do see wind and snow, and there the last two rows are the ones that matter. Against the same board laid flat, which is how a lot of framing ends up loaded, the wedge is far stiffer.

The comparison in full.

One consolation, before the argument

There is a caveat to the caveat, and it is a fair one.

The comparison above puts the 2x4 **on edge** — the orientation in which a board is strongest, and the way a carefully framed floor joist is installed. A great deal of real framing does not get that. Sheathing battens, blocking, purlins and plenty of cheerfully improvised structures end up loading boards the flat way, and against a 2x4 laid flat the wedge is **5.4 times stiffer**.

So the honest summary is: worse than a well-oriented board, dramatically better than a badly-oriented one.

That is still not the argument.

Which is the right question

The argument is that *the comparison itself is the wrong shape*.

Asking “is this wedge as good a beam as a 2x4?” assumes that being a beam is the job. In a rectangular building it usually is: a stud, a joist and a rafter all spend their lives resisting bending, largely alone.

A triangulated frame does not work that way. That is the entire point of triangulating it. A properly designed triangle converts an external load into forces running *along* its members — compression down one, tension along another — instead of asking any single member to bend. The network carries the load; the sticks mostly just push and pull.

And for a member being pushed or pulled along its length, the property that matters is **cross-sectional area**. Not section modulus. Area.

Which is the column the wedge wins, by +20 per cent.

So the useful question is not:

Is this wedge as good a rectangular beam as a 2x4?

but:

Is this wedge an effective axial member inside this triangulated system?

And there the answer is straightforwardly yes, with more wood in the section than the board it replaces, from a tree that never went near a mill.

Mostly

One more piece of honesty, because “mostly axial” is doing real work in that paragraph.

Panels are not purely axial. Wind pushes on the skin between the members and snow sits on it, and those loads reach the frame as bending in the individual sticks. It is not the dominant case and it is not nothing, and it is exactly where the wedge’s -33 per cent deficit is a real deficit rather than an academic one.

Two things about that:

- The wedge is deep. 4.00 inches pith to bark is more depth than the board has, and depth is what resists the bending it does see — which is why the stiffness came out level rather than behind despite the shape being worse.
- Panel span is short. The distance between supports in a 2V panel is a few feet, not the fifteen a floor joist crosses, and bending demand falls away fast as spans shorten.

None of which is a substitute for somebody competent checking your particular building for your particular snow load. Chapter 10 says the same thing at more length and names who to ask.

What this chapter does not claim

It is worth fencing the argument, so nobody quotes it doing work it cannot do.

This chapter does **not** claim that a split-log wedge is better than a 2x4 as a wall stud, a floor joist, a roof rafter, a shelf, a header or a piece of furniture. For all of those the board wins, often easily, and the reasons are in the table above.

It claims something narrower and much more defensible:

For a purpose-designed geodesic dome whose geometry, connectors and jigs are built around radially split timber, the wedge is a better use of the tree than first converting that same tree into ordinary dimensional lumber.

That is the whole claim. It depends on the building being designed around the material from the beginning — which is what the rest of this book is about.

We are not asking an unconventional piece of wood to imitate a 2x4.

We are building something in which it does not have to.

Forty Panels, One Hundred and Twenty Members

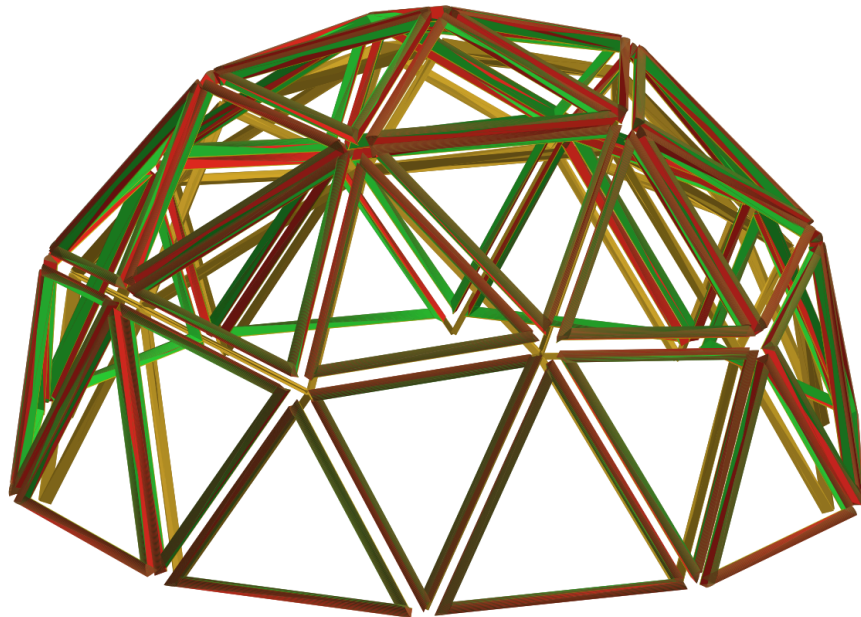
The $2V$ hemisphere, counted

This chapter is planned but not written yet. Its page plan is in the outline.

Forty Panels

Where $2V$ comes from

The frame, exploded



Forty panels, 120 members, 55 seams between them.

What the frame contains

What the frame contains

	count
Panels (triangles)	40
Members (struts)	120
Members per panel	3
Unique edges	65
Seams between two panels	55
Rim edges, joining nothing	10

Counted from the solved dome, not chosen. The distinction between edges and seams matters: you close 55 joints, not 65.

Counted from the solved dome, not chosen.

Why hemispheres and not spheres

Two Lengths, Not Forty

The whole reason a dome can be built by one person

This chapter is planned but not written yet. Its page plan is in the outline.

Two Lengths

Short and long

The member schedule

Every stick in the dome

class	count	each (in)	each (ft) / all (ft)
SHORT	30	62.731	5.228
SHORT	30	63.267	5.272
LONG	30	71.241	5.937
LONG	30	72.000	6.000
total	120		673.1

Two edge classes become four lengths because the pinwheel inset differs at each end. Cut to these and the dome closes. The last column is each member's length in feet, except on the total row, where it is all of them added together.

Every stick in the dome, in four lengths.

Why the pinwheel splits two into four

Compression, Tension, and One Eighth of a Tree

What a wedge strut can actually carry

This chapter is planned but not written yet. Its page plan is in the outline.

One Eighth of a Tree

Along the grain, and across it

The end-to-side bearing

What this book will not tell you

The number I will give you

What a Dome Costs You

The chapter that argues against the book

This chapter is planned but not written yet. Its page plan is in the outline.

What a Dome Costs You

Furniture does not fit a curve

Roofing a compound surface

Permits, appraisals and resale

When to build a rectangle

PART 3

The Wedge

“Same stick, four ways up.”

The paradigm itself: how a split trunk becomes a structural member, and the four things it can do.

Split Like a Cake

Three passes, eight sectors, and why eight

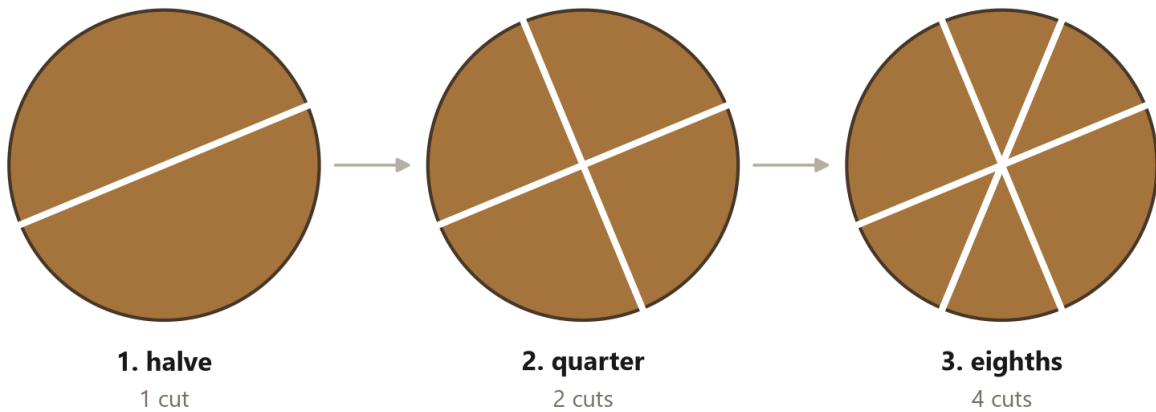
This chapter is planned but not written yet. Its page plan is in the outline.

Split Like a Cake

The three passes

Three cuts, eight sticks

Three passes, in this order



The halving pass goes first every time: it is the only cut with a flat reference on both sides afterwards. Cut the eighths first and the last two sectors have nothing to hold them.

The order of the passes. Halve, quarter, eighth.

Why eight and not six or twelve

What other split counts give you

Splitting a 10-inch section other ways

splits	angle	width in	depth in	sq in	= 2x4s
4	90	7.07	5.00	19.63	2.45
6	60	5.00	5.00	13.09	1.64
8	45	3.83	5.00	9.82	1.23
10	36	3.09	5.00	7.85	0.98
12	30	2.59	5.00	6.54	0.82
16	22	1.95	5.00	4.91	0.61

Depth does not change with the split count -- it is always pith to bark. Only the width does. Eight is the book's choice: a sixth is too heavy to handle alone and a twelfth is too slender to bear on its own end.

Sector width and depth against split count.

Eighty-Seven Percent

The recovery arithmetic, with the kerf paid for

This chapter is planned but not written yet. Its page plan is in the outline.

This chapter corrects something. The 88% recovery figure in the project brief, which was computed for a 15-inch butt rather than this book's 12-inch tree.

Eighty-Seven Percent

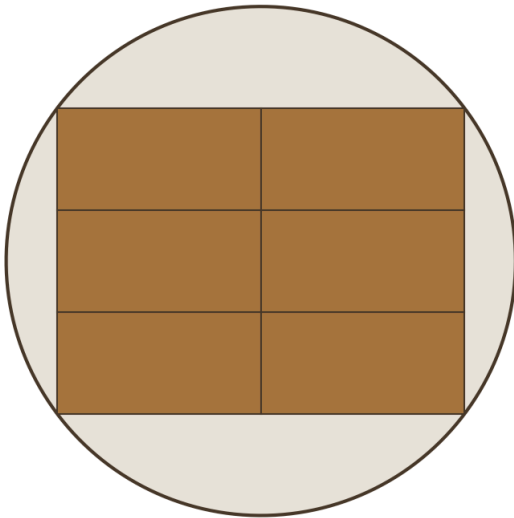
What a mill loses, and to what

What the wedge loses

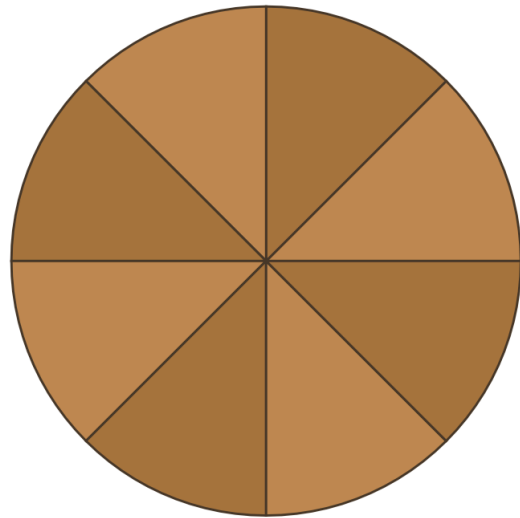
Both conversions, same log

The same log, converted two ways

x1.93 the usable wood



6 boards, 45%



8 members, 87%

Left: grid-packed into 2x4s the way a mill saws, sized on the small end -- 45% of the solid wood. Right: split into 8 sectors, losing only the kerf -- 87%. The shaded area on the left is slabs, edgings and trim.

The same log, converted twice. 87.4 percent against 45.

Section by section

Every bucked section, converted both ways

sec	butt in	top in	solid bf	kerf bf	wedge bf	2x4s
1	15.0	13.1	155.3	14.1	141.2	11
2	13.1	11.2	116.2	12.1	104.0	7
3	11.2	9.3	82.8	10.2	72.5	4
4	9.3	7.4	55.0	8.3	46.6	2
5	7.4	5.5	32.9	6.4	26.5	1
all			442.1	51.2	390.9	25

The 2x4 column is the same round section grid-packed the way a mill would saw it, sized on its small end because a board has to be full width along its whole length.

Every bucked section, both conversions.

Where 88 became 87

Four Ways Up

One stick, four orientations, four different buildings

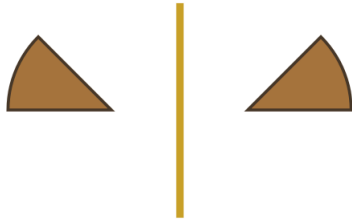
This chapter is planned but not written yet. Its page plan is in the outline.

Four Ways Up

Rotating about its own axis

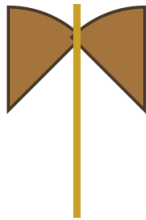
All four, side by side

The same stick, four ways up



point dome in

both points inward, toward the centre of the dome



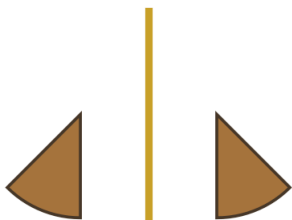
point panel in

points apart, each into the middle of its own triangle



point dome out

both points outward, away from the centre, at the sky



point panel out

points toward each other, across the seam

Each pane is one seam seen end-on: two members, one from each panel, with the seam between them in gold. Only the rotation of the members about their own long axes differs -- the sticks themselves are identical. The reading under each pane is the simulator's own, so the book and the tool cannot describe the same rotation differently.

The same pair of sticks, four ways up.

Point dome in

Point panel in

Point dome out

Point panel out

Choosing one

The four orientations, at a seam

orientation	what a pair of points does
point dome in	both points inward, toward the centre of the dome
point panel in	points apart, each into the middle of its own triangle
point dome out	both points outward, away from the centre, at the sky
point panel out	points toward each other, across the seam

Read straight from the simulator's own seam readings, so the book and the tool cannot describe the same rotation differently.

Read straight from the simulator's own seam readings.

The Pinwheel

Three sticks, one triangle, and not one mitre

This chapter is planned but not written yet. Its page plan is in the outline.

This chapter corrects something. The ‘no mitres anywhere’ claim, already corrected on camera in lesson_wedge_why.

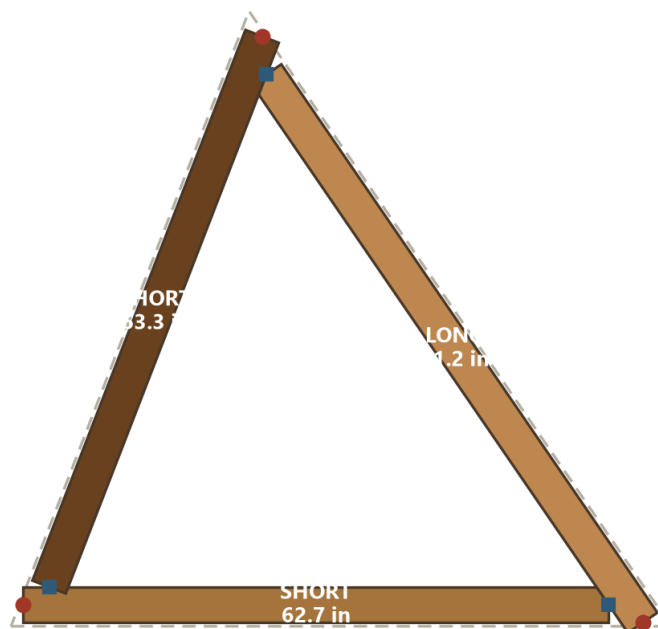
The Pinwheel

Why mitres were the enemy

Each end lands on a side, not a point

One panel, taken apart

One panel: three sticks, pinwheeled



circle = tail, receives the member behind it (4.1 in of bearing)
square = head, lands on the next member's side

Each member's end lands on the SIDE of the next one, never on its point, so no end is ever mitred to another end. The dashed outline is the true triangle; the members sit inside it by the inset. Drawn from the rectangular-band model, where both ends come out square. Split a real log sector and the tail becomes a compound cut -- a mitre plus a bevel of half the sector angle. Chapter 31 is about that cut and does not pretend it is square.

Three members, three end-to-side joints, no mitre anywhere.

What the inset costs

The mitre claim, corrected

The Seam and Its Key

Where two panels meet, and the angle nobody wants to hear about

This chapter is planned but not written yet. Its page plan is in the outline.

The Seam and Its Key

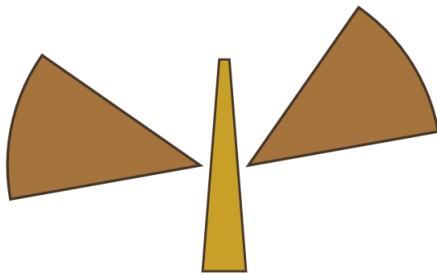
The fold angle is not constant

Raw trapezoid, or shaved flat

One seam, both ways

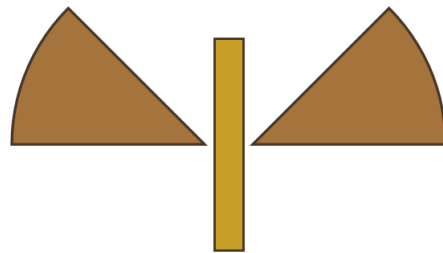
One seam, closed two ways

raw trapezoid



key is tapered, one per seam

shaved flat

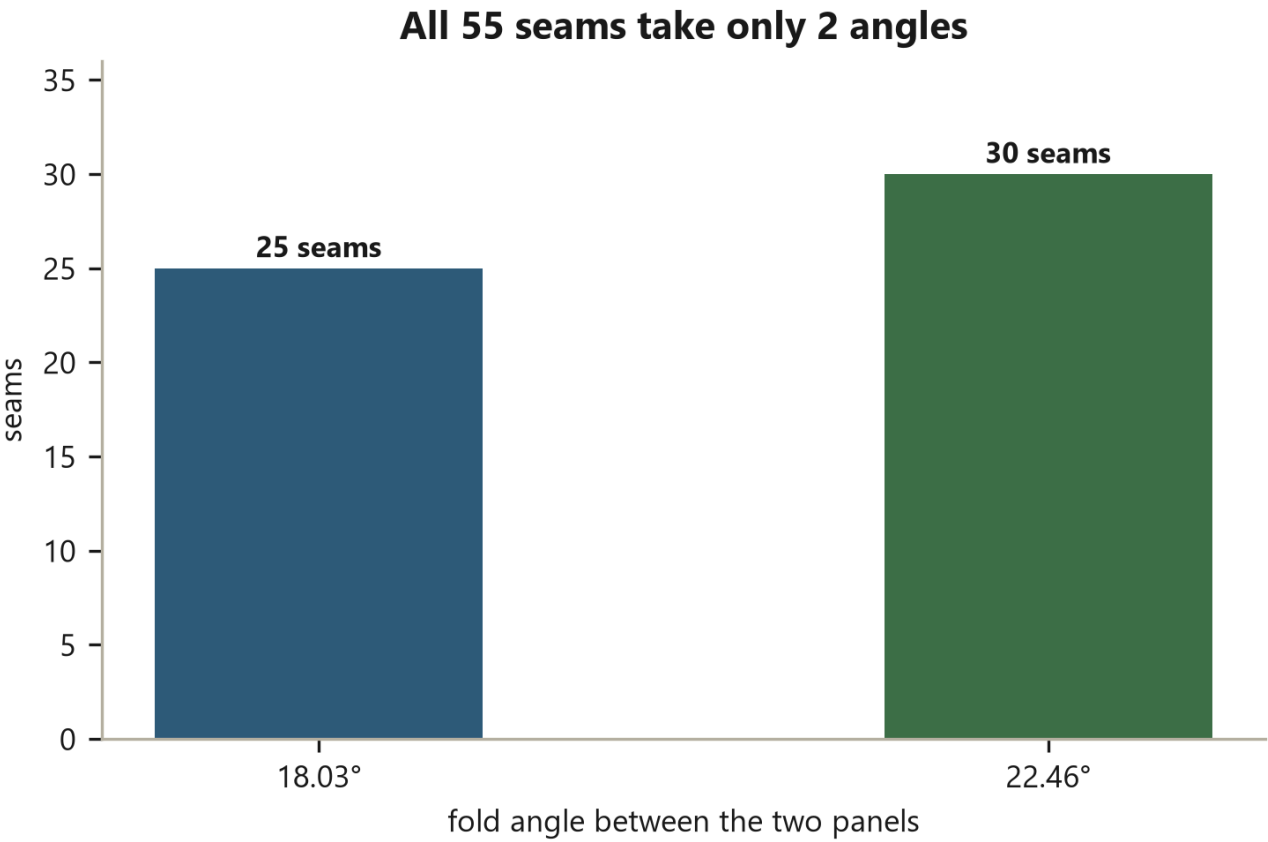


key is rectangular, one for all

The fold angle across this dome runs from 157.54 to 161.97 degrees. Left: leave both split faces exactly as sawn and let a tapered key take up the difference -- no machining, but every key is its own shape. Right: plane both faces parallel and one rectangular key fits everywhere -- one extra pass per stick.

Leave the split faces alone and taper the key, or plane the faces and use one key everywhere.

Every seam in the dome



This is better news than it looks. The seams are not all alike -- but they are not all different either. 2 fold angles across 55 seams means 2 key profiles, cut once and repeated. The raw gap the key has to fill runs 22.54° to 26.97°.

All 55 seams and their fold angles.

Hose, key or nothing

Let the Connector Hold the Angle

Keep the wood, machine the small part

Let the Connector Hold the Angle

There is a principle underneath this whole method, and it is worth stating on its own because it is the thing that makes rough natural timber buildable to a tolerance at all.

The tree supplies the mass. The jig supplies the repeatability. The connector supplies the precision.

Three jobs. Three different things doing them. Nothing asked to do two.

Precision is expensive per cubic inch

The instinct with an irregular material is to machine it until it is regular. Plane the faces, square the ends, bring everything to a dimension, and then build with confidence.

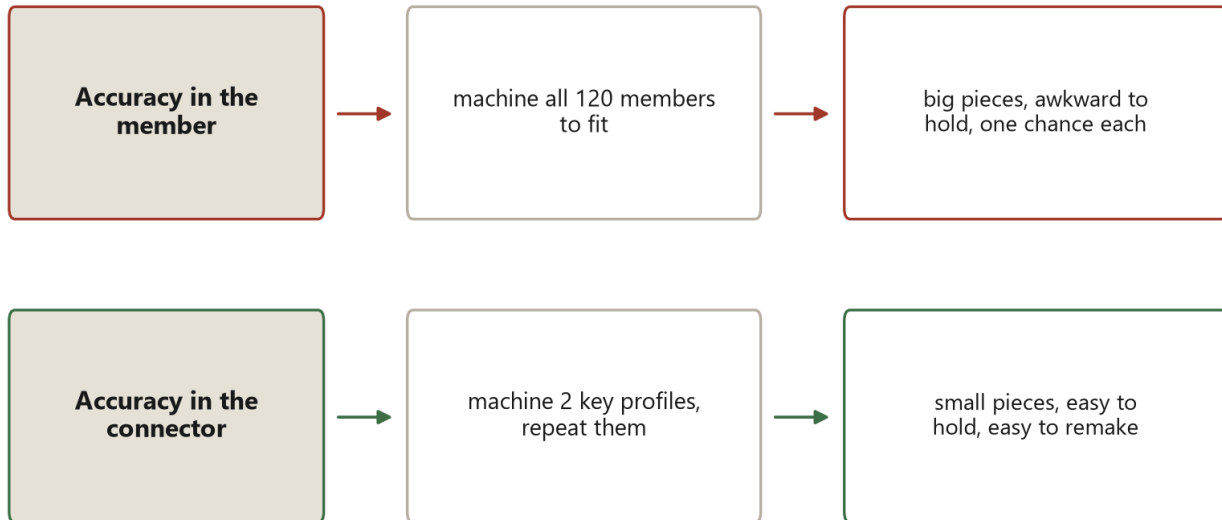
That instinct is why dimensional lumber exists, and inside a mill it is the right instinct — a mill can afford to hold a tolerance because it is holding it on thousands of pieces with a machine built for nothing else.

A person with a chainsaw cannot afford it, and the reason is not skill. It is that **precision costs roughly in proportion to how much material you have to be precise about**. A long, heavy, awkward member is expensive to hold, to reference, to feed and to correct. A small block is cheap to hold and cheap to remake if you get it wrong.

So there is a choice about *where to put the accuracy*, and the two answers lead to very different afternoons:

Where the precision lives

The tree supplies the mass. The jig supplies the repeatability. The connector supplies the precision.



Both routes end with a dome that closes. One asks for accuracy 120 times, on the largest and most awkward pieces in the build. The other asks for it 2 times, on the smallest, then repeats the result 55 times. This is not a way of avoiding accuracy. It is putting it where it is cheapest to buy.

Where the precision lives.

One route asks for accuracy 120 times, on the largest and most awkward pieces in the build, with one chance at each. The other asks for it 2 times, on the smallest pieces, and then repeats the result 55 times.

Both end with a dome that closes.

This is not a way of avoiding accuracy. Nothing here is less accurate than the alternative — the seams end up at the same angles either way. It is a question of *where the accuracy is cheapest to buy*, and the answer is: in the small part.

The wedge starts close to useful

There is a piece of luck in this, and it is the reason the wedge suits a dome rather than merely tolerating one.

A rectangular board arrives with two faces at ninety degrees to each other. Ninety degrees appears nowhere in a geodesic dome. Every angular relationship in the frame has to be created from nothing, by cutting.

A split sector does not arrive square. It arrives with two sawn faces that each sit **22.5 degrees** off the member's own centreline, because that is half of one 45-degree sector.

And the seams of this dome fold by 18.03 and 22.46 degrees.

Those are not the same numbers. The wedge does not magically arrive at the right angle — that would be too good, and this book does not do too good. But it arrives *within a few degrees* of what the joint wants, rather than sixty-odd degrees away.

The natural shape of the tree becomes part of the joint system instead of an obstacle that first has to be removed. That is the difference between a material that suits a structure and one that merely survives it.

Two ways to close the difference

The gap between the face you have and the face the joint wants can be closed from either side.

Shape the key. Leave both sawn faces exactly as split and cut a tapered key that fills whatever is left. No wood is machined at all. The cost is that the key is not the same shape at every seam, so there is more than one key profile to make.

Shape the wood. Plane both faces parallel and drop in a plain rectangular key that fits everywhere. One key profile for the whole dome. The cost is a machining pass on every stick.

Chapter 16 covers what each does to the seam. This chapter is about what the second one costs, because it sounds much worse than it is.

What planing the faces actually costs

Here is the arithmetic, and it is the most reassuring table in the book.

The raw split face sits at 22.5 degrees. A flat key wants a face at half the seam’s fold angle. The difference between those two is everything that has to come off.

What the shaved-flat option costs, over a 2-inch mating land

key	seams	fold	half fold	raw face	correct	cut in	of depth
K1	25	18.029	9.014	22.5	13.486	0.480	9.6%
K2	30	22.459	11.229	22.5	11.271	0.399	8.0%

The raw split face already sits at 22.5 degrees off the member's own centreline, and a flat key wants half the seam's fold angle. The difference is all that has to come off -- under half an inch, on a member 5.00 inches deep. That is the whole price of the option that makes one key profile fit everywhere.

What the shaved-flat option costs.

Over a two-inch mating land — you do not need the whole face to bear, only a band of it — the deepest cut anywhere in the dome is **0.48 inches**.

On a member 5.00 inches deep, that is 10 per cent of the section.

So even the “machine the wood” option does not convert a wedge into something resembling dimensional lumber. It removes a narrow sliver from a narrow band and leaves the overwhelming majority of the member exactly as the saw left it. You are still not making boards. You are making one precise interface on an otherwise raw stick.

That is worth knowing before you decide, because the shaved-flat route is often written off as a betrayal of the whole idea. It is not. It is a 10 per cent concession that buys you a single key profile.

Or do not machine anything at all

There is a third answer, and in some builds it is the best one: make the connector out of something that does not need to fit exactly, because it deforms.

A rubber or closed-cell spline, a length of hose, a compressed gasket — any of these fills a *range* rather than a value. Feed it a seam anywhere between the two fold angles and it takes up the difference by squashing.

What a compliant key can do, all at once:

- absorb the angular difference,
- absorb manufacturing variation in the members,
- seal the seam against weather,
- isolate one panel from another acoustically and against vibration,
- hold a preload so the joint stays tight as the wood moves.

What it cannot do is carry the load a timber key carries. A compliant seam is a weather joint and a tolerance joint; it is not a structural one. So the choice depends on a question you should answer before you buy anything: **is this seam holding the building up, or holding the weather out?**

In a frame where each panel is a complete, self-braced triangle — which is what Chapter 18 is about — a great deal of the shell's stiffness lives inside the panels rather than across the seams, and that is what makes the compliant option viable at all.

Why this is a manufacturing idea, not a dodge

It is worth defending this properly, because “let the connector take up the difference” can sound like an excuse for sloppy work.

It is the opposite. It is how repeatable production works everywhere.

A machine shop does not hold a tolerance on every surface of every part; it holds it on the surfaces that locate and mate, and leaves the rest rough because rough is cheaper and does no harm. An engine

block is machined to a few thousandths where the bearings sit and left as cast everywhere else. Cabinetmakers scribe one edge to a wall and leave the other three alone.

The principle is always the same: **decide which surfaces do work, and spend your accuracy there.**

In this dome, the surfaces that do work are the mating faces of the seam and the ends where one member bears on another's side. Everything else — the bark, the taper, the natural bend, the exact diameter — can vary, and the jig is designed so that it can.

The tree is not being asked to be accurate. It is being asked to be strong, which it already is.

Why Every Panel Keeps Its Own Edge

One hundred and twenty members for sixty-five edges, and what the difference buys

One hundred and twenty members for sixty-five edges, and what the difference buys

Every Panel Keeps Its Own Edge

If you have been adding up as you read, something has not been adding up.

The arithmetic that does not look right

The frame has 40 triangular panels. Each panel has 3 members. So:

$$40 \times 3 = 120 \text{ members}$$

But the shell only has **65 edges**. Count the lines on a drawing of the dome and that is what you get: 55 interior edges where two panels meet, plus 10 around the open bottom where a panel meets nothing.

So there are 55 more members than there are edges for them to be.

That is not an error. It is the single biggest deliberate decision in this frame, and it costs **1.85 times** the timber a conventional geodesic skeleton would use.

It is worth being clear about the size of that, because it is not a rounding difference. If you built the same dome with one shared strut per edge you would need 65 members instead of 120. The panelised frame uses almost twice as much wood in the frame.

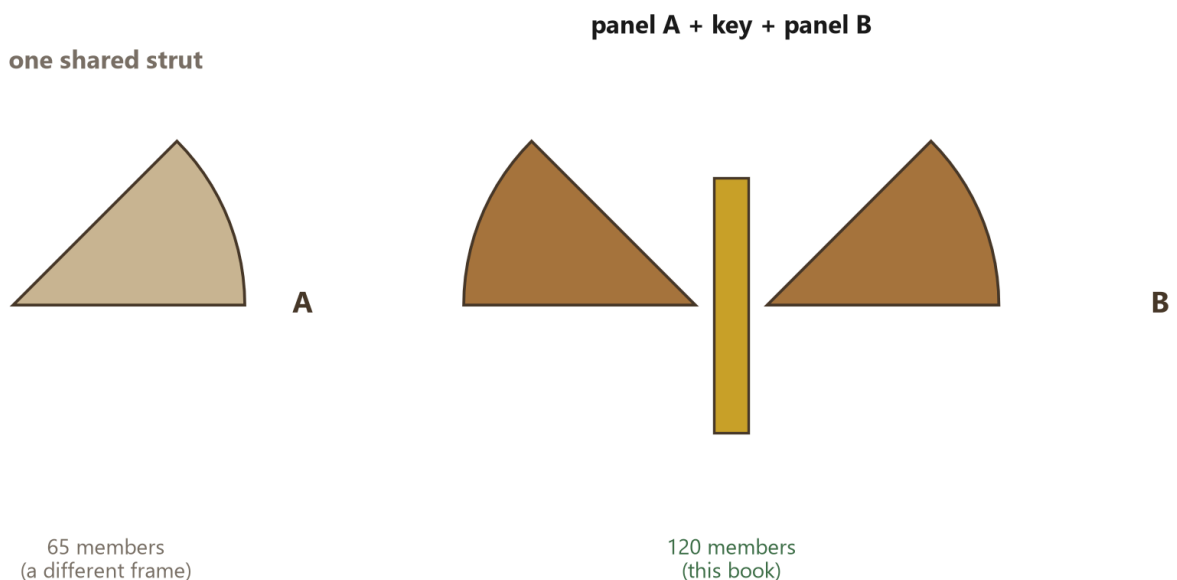
Here is why it is worth it anyway.

What is actually at a seam

In a classical geodesic frame, an edge is one strut, and the two triangles either side of it both fasten to that same strut. Efficient, and the standard way it is done.

In this frame, each panel is built complete — three of its own members, closed into its own triangle — before it goes anywhere near its neighbours. So where two panels meet there are two members, one belonging to each, with the key between them.

One seam, two members



40 panels x 3 members is 120, but the shell has only 65 edges: 55 members exist twice over, once for each panel meeting there. That is 1.85 times the timber a shared-strut frame would use, and it is what lets every panel be built flat, checked, skinned and lifted as a finished thing.

One seam, two members.

Panel A's member. The key. Panel B's member.

The duplicated edge is not lumber somebody forgot to remove. It is the price of a panel being a *thing* rather than a position in an assembly.

What the extra wood buys

Six things, and the first three are why the fortnight in Part 5 is a fortnight rather than a season.

A panel can be built flat. All three of its members exist and belong to it, so the whole triangle can be assembled on a bench, at waist height, on the ground, in the dry. Nothing about building it requires any other panel to exist yet. In a shared-strut frame you cannot finish a triangle without the struts its neighbours also use, which is why those domes tend to get assembled in the air, strut by strut, at the top of a ladder.

A panel can be checked before it goes up. It has a measurable shape while it is still somewhere you can measure it. A panel that came out wrong is firewood and half an hour; a panel discovered wrong when it is the thirty-eighth thing lifted into place is a much longer day.

A panel can be finished before it goes up. Skin it, seal it, insulate it, run a cable through it, paint it — all on the ground. This is a genuine transformation of the work: it turns awkward three-dimensional construction at height into repetitive two-dimensional production at a bench, which is manufacturing rather than improvisation.

The seam gets room for a gasket. Two members with a gap between them is a place a key, a spline or a compressible seal can live. One shared strut has no such space — the panels either side bear directly on it, and weatherproofing that joint is a separate problem solved with tape and hope. Chapter 17 is about what can be done with that space, and Chapter 16 about what has to be.

A panel can be taken out again. Undo one seam's fasteners and a panel lifts out, leaving its neighbours standing and complete. Nothing else in the shell depended on that panel's members. Replace a rotted one in year fifteen, cut a window into one in year three, take three out to get a bathtub in — these become ordinary jobs rather than partial demolitions.

The frame gains redundancy. Every interior edge has two members in it. One splitting, checking or being damaged does not leave that edge unsupported. It is not a designed redundancy and nobody should rely on it, but it is real.

The check that has to close

Two independent ways of counting the members have to give the same answer, and this is worth doing yourself once, because it is the check that tells you the frame in your head is the frame on the ground.

Count panel by panel. Every panel has 3 members.

$$40 \times 3 = 120$$

Count edge by edge. Every interior seam holds two members, one from each panel. Every rim edge holds one, because there is nothing on the other side of it.

$$(55 \times 2) + 10 = 120$$

Both give 120. The software in this project runs exactly this check every time the book is built, and refuses to print anything if the two disagree — because if they ever did, one of the two pictures would be of a building that does not exist, and there would be no way to know which.

When to share a strut instead

This chapter should not pretend the choice is obvious in every case.

Share the strut if:

- **Timber is your scarcest resource.** 1.85 times the members is 1.85 times the trees, and if you have one tree rather than two that decides it.
- **You have a crew and scaffolding.** The panelised method's main advantage is that one person can do it on the ground. With four people and a lift, in-air assembly is not the ordeal it is alone.
- **The dome is small.** Below a certain size the members are short enough to handle comfortably at height, and the panel machinery is overhead you do not need.
- **You want the lightest possible frame.** For a temporary structure, a greenhouse, or anything that has to be carried in, the weight of the duplicated members is a real cost.

Keep the panel if:

- **You are building alone,** which is what this method was designed for.
- **You want to finish panels on the ground,** which is most of the reason the build fits in a fortnight.
- **You want a weatherable seam,** because the gap is where the seal goes.
- **You expect the building to change,** because panels come out.

This book chose the panel, and every number in it follows from that choice. Chapter 22's two trees are two trees precisely *because* the frame duplicates its edges. A shared-strut dome of the same diameter would come out of about one.

That is the trade, stated plainly: **roughly twice the wood, for a building one person can make on the ground and take apart again.**

Bends, Knots and Small Units

Why six-foot pieces forgive what sixteen-foot boards do not

This chapter is planned but not written yet. Its page plan is in the outline.

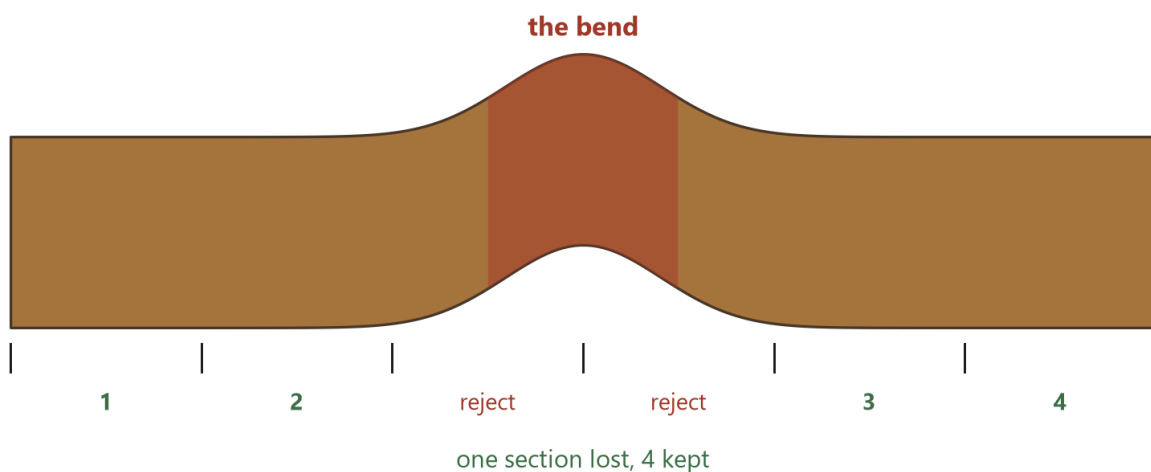
Small Units

A bend costs you a section, not a tree

Knots, and where they are allowed

One bent trunk, salvaged

A bend costs a section, not a tree



A mill wants a straight log: the bend spoils the whole length because a board has to be straight along all of it. Bucked into 6-foot pieces, the bend costs one section. Cut it out and carry on either side of it.

A bend a mill would refuse, and the sections either side of it.

Grade your own stock

PART 4

The Two Calculations

“Either you know the dome you want, or you know the tree you have. Never both.”

The manual's core. Two methods, run in opposite directions, and the proof that they close on the same dome.

Which Way Round Are You Working?

Choose your method before you touch anything

Which Way Round

There are two ways to size a wedge dome, and the first thing you have to do is work out which one you are actually in. Not which one you would like to be in. Which one you are in.

If you get this wrong you will spend a week doing arithmetic that describes a building you cannot build, and you will not find out until the day you go to cut, when the longest stick the frame wants turns out to be eight inches longer than the longest stick you own.

Everything in the next five chapters follows from this one decision, so it gets its own short chapter and no other content.

Design first, or tree first

Design first is the way everybody expects to work. You know what you want: a room of a certain size, a ceiling you can stand under, a footprint that fits between the trees you are not cutting down. You start from the dome and you work backwards to the woodpile. The output is a shopping list — this many sticks, this long, and therefore this much trunk.

Tree first is the way this book was actually written. You have trees. They are the trees you have: a certain height, a certain taper, a certain amount of straight length between the butt flare and the first serious bend. You start from what a section of that trunk can be cut into and you work forwards to the dome. The output is not a shopping list. It is a diameter, and you find it out rather than choosing it.

Both are honest. Neither is better. They answer different questions and they run in opposite directions, and the entire content of Chapters 21 and 22 is those two directions worked through in full.

The honest test

Here is the test, and it is uncomfortable.

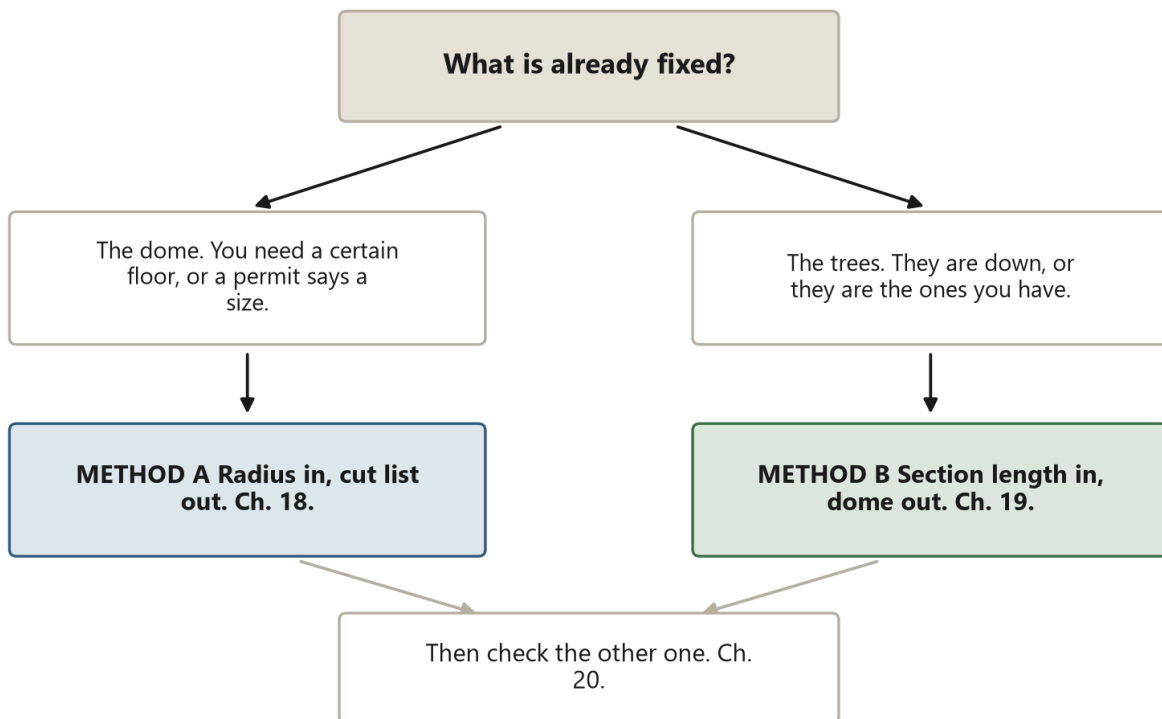
Are the trees already down, or already chosen?

If they are — if you are standing next to a windfall, or you have marked two pines at the back of the property, or somebody has offered you a load — then you are in Method B, and picking a diameter is not a decision you get to make. It is a wish. You may still write a number on a piece of paper. The trees will not read it.

If the trees are genuinely not chosen — you will buy or fell whatever the design calls for, and there is enough standing timber that a foot either way on the bucking length costs you nothing — then you are in Method A, and you should design the dome you want.

Almost everybody thinks they are in Method A. Most people are in Method B. The tell is whether you would actually go and get a different tree if the arithmetic asked for one. If the answer is “well, I’d probably make it work with these,” you are in Method B and you may as well admit it on page one rather than on cutting day.

Which method are you in?



Almost everybody thinks they are designing first. If the trees are already down, or already chosen, you are in Method B and choosing a diameter is a wish, not a decision.

Start from what you actually have.

What both methods share

Underneath the two directions there is one relationship, and it is worth understanding before you use either method, because it is the thing that makes both of them exact rather than approximate.

In a 2V dome of radius R , every panel is a triangle whose corners sit on the sphere. If the members simply ran corner to corner you could scale them straight off the radius: double the dome, double the stick. But they do not run corner to corner. They pinwheel — each member's end lands on the *side* of the next one rather than on a shared point — and the amount each member is set back from the true corner depends on **how wide the member is**, not on how big the dome is.

A wider stick sits back further. And the width of your stick comes from the diameter of your log, which has nothing whatsoever to do with the dome.

So member length is not proportional to radius. It is radius *minus* a correction that stays the same size while the dome grows. Which means:

- Going from radius to member length is a calculation.
- Going from member length back to radius is that same calculation, solved backwards.

Neither direction is a rule of thumb, a table, or a fudge factor. They are one function and its inverse, and Chapter 23 exists solely to run them against each other and print the difference, which is 0.0e+00 inches.

Before you go on

Whichever method you are in, you need three measurements before either chapter is any use to you:

1. **The diameter of your log at the middle of a section.** Not at the butt — the butt is the fattest part and it flatters you. The middle, because that is what the average stick will be.
2. **The straight length you can get out of the trunk.** Between the flare and the first bend that would spoil a section.
3. **The longest piece you are willing to carry on your own.** This is a real constraint and it does more to set the dome's size than anything you will read in a book. A six-foot green pine wedge is heavy. An eight-foot one is a two-person lift, and if there is only one of you, that is the end of the discussion.

Write those three numbers down. Then turn to Chapter 21 if you are designing first, or Chapter 22 if the trees are already on the ground.

And then, whichever one you used, read Chapter 23 and check it with the other one. That takes ten minutes and it is the cheapest insurance in the book.

Method A: From the Dome You Want

Radius in, cut list out

Method A

You know what you want. Now find out what it costs in trunk.

This is the method to use when the timber is genuinely not decided — when you will fell, buy or scrounge whatever the design asks for. It runs from the building to the woodpile, which is the direction every other construction book works in, and it is the more comfortable of the two.

It is also the one that will tell you, politely and in advance, when the dome you have in mind needs more trees than you have.

The procedure

1. Start from floor, not radius

Nobody wants a radius. They want a room.

Decide the floor area you want to stand on and convert:

$$radius (feet) = \sqrt{(floor \div \pi)}$$

300 square feet gives a radius of 9.77 feet — call it 117 inches. A 2V hemisphere's standing height at the centre is its radius, so that same number is also your ceiling.

This is the first place a dome surprises people. A 300-square-foot dome is 9.77 feet tall in the middle, which is more house than the floor area suggests, and the volume is doing something a rectangle of the same footprint cannot.

It is also the first place a dome disappoints people, because not all of that floor has headroom over it. Chapter 43 has the map.

2. Measure the stock you expect to have

Even in Method A you cannot skip this. The pinwheel setback depends on how wide your members are, and you cannot compute a member length without it.

If you have not felled anything yet, use the diameter you expect to be cutting. A 12-inch butt tapering to 8 gives sectors about 3.83 inches wide at mid-trunk, and that is the figure this book's tables assume.

If your logs will be fatter, your members will be wider, your setback will be larger, and every length in your cut list will come out shorter than the tables here. The tables are computed at this book's stock; recompute for yours.

3. Solve the member lengths

For a chosen radius, the layout gives you 4 member classes. This is the calculation Chapter 20 described, run in the forward direction: radius in, lengths out.

At 300 square feet the longest member comes out at 64.5 inches and the shortest at 56.1.

Method A, worked: 300 square feet

step	value	where it comes from
Floor you want	300 sq ft	chosen
Radius that gives it	117.26 in	$\sqrt{\text{area}/\pi}$
Dome diameter	19.54 ft	2 x radius
Standing height at centre	9.77 ft	= radius
Your stock's sector width	3.83 in	10 in log, 8 ways
Longest member	64.55 in	solved
Shortest member	56.14 in	solved
Buck to	6 ft	longest member, rounded up
Sections needed	15	$120 / 8$
Trunk feet needed	90 ft	sections x bucking length
Trees needed	1.88	at 48 usable ft each
So fell	2	rounded up

Every line is computed from the line above it. Change the floor area and every number below it moves.

300 square feet, worked all the way through.

4. Round the bucking length up

Your longest member is 64.5 inches. Buck to 6 feet — 72 inches.

You cannot buck to 64.5 inches. Well, you can, but you should not: a section cut to exactly the longest member leaves no allowance for the head-end overfit (Chapter 35), no allowance for the butt cut (Chapter 34), and nothing at all for the end of a log being split or shaken.

Round up to the next whole foot and take the waste. It is the cheapest line item in the build.

This does mean the design-first path throws away more wood than the tree-first path, because the tree-first path chooses a dome that uses the whole section. That is a real cost of getting to pick your diameter, and it is worth saying out loud rather than discovering in the offcut pile.

5. Count sections, then trunk, then trees

The frame needs 120 members. Each section gives 8, so:

$$\text{sections} = 120 \div 8 = 15$$

$$\text{trunk needed} = 15 \times 6 \text{ ft} = 90 \text{ feet}$$

$$\text{trees} = 90 \div 48 = 1.88$$

So: **fell 2.**

Notice the number. Designing a 300-square-foot dome from scratch and sizing the trees to it lands on the same 2 trees that this book's actual build used, from the opposite direction. That is not a coincidence — it is the size of dome a pair of ordinary pines makes, and it is why this method produces houses of roughly this size over and over.

6. Check it against the tree you can actually get

Then go and look at the trees.

If the tallest straight trunk you can find is thirty feet rather than 48, your 1.88 trees becomes three. If your logs are much thinner at the butt than 12 inches, your members are narrower, your sticks are longer for the same dome — and thinner, which Chapter 10 has opinions about.

This is where Method A hands over to Method B, and where Chapter 23's round trip earns its place.

Where this method surprises people

Member length does not scale with the dome. Double the radius and the members do not double. They grow by slightly less, because the pinwheel setback stays the same size while the triangle grows around it.

This is good news at large sizes and bad news at small ones. A very small dome has members so short relative to their own width that the setback eats a serious fraction of them, and below a certain size the geometry stops making sense at all — a stick that is nearly as wide as it is long is not a strut, it is a block. Chapter 49 says where that line is.

Height is not a free variable. In a plain hemisphere, height equals radius. If you want a wide floor and a ceiling taller than half that width, you cannot have both from a hemisphere; you need a riser wall under it (Chapter 41) or a different fraction of a sphere, which is a different book.

The trees number is rarely a whole one. You will get 1.88, or 2.4, or 3.1. Round up and accept the leftover, or shorten the bucking length and see whether the arithmetic lands better. There is no third option that gets you a partial tree.

The lookup table

If you would rather not do any of this, the table below is the whole method precomputed at this book's stock section. Find your floor area on the left and read across.

Design first: pick a floor, read the rest

floor sqft	diam ft	height ft	longest in	buck ft	trees
80	10.1	5.0	29.5	3	0.94
120	12.4	6.2	37.9	4	1.25
180	15.1	7.6	48.2	5	1.56
250	17.8	8.9	58.2	5	1.56
320	20.2	10.1	66.9	6	1.88
400	22.6	11.3	75.8	7	2.19
500	25.2	12.6	85.6	8	2.50
650	28.8	14.4	98.8	9	2.81

Computed at this book's stock: a 10-inch log split 8 ways. A fatter log gives wider members, which changes the longest-member column and therefore everything after it.

Floor area to trees.

Two warnings about using it:

1. It assumes members 3.83 inches wide. Different logs, different table.

2. The `trees` column assumes 48 feet of usable trunk per tree. If your trees are shorter — and most are — scale it.

Then check it the other way

You now have a cut list. Before you cut anything, take the longest member off that list, hand it to Method B, and see what dome comes back.

It should be the dome you started with. If it is not, you have made an arithmetic slip or changed an assumption halfway through — a different gasket thickness, a different split count, a different stock width.

That check is Chapter 23, it takes ten minutes, and it is the reason this book has two methods rather than one method and a warning.

Method B: From the Tree You Have

Section length in, dome out

Method B

This is the method the book is named after, and it is the one that produced the dome in the photographs. It starts from a tree lying on the ground and ends with a number you did not choose.

That last part is the whole character of the method. In Method A you decide how big the building is and the woodpile has to comply. Here the woodpile decides, and your job is to find out what it decided.

The procedure

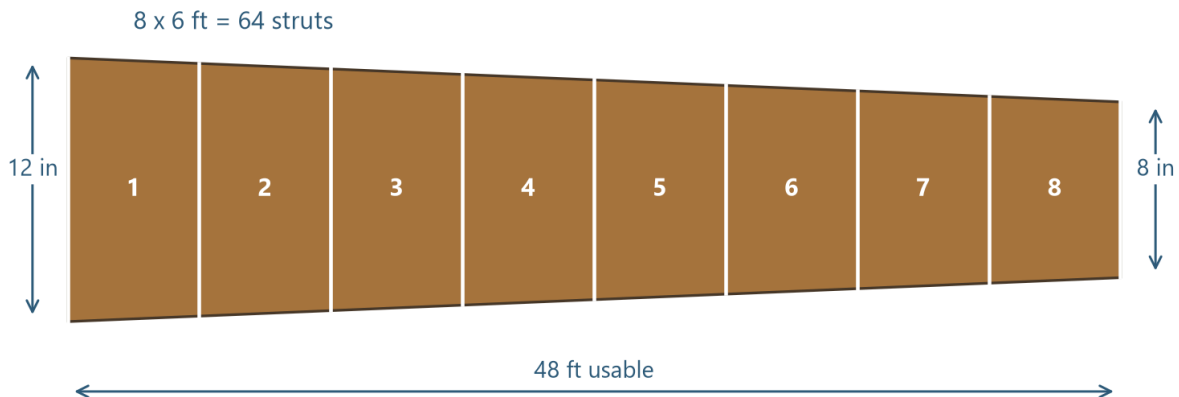
1. Measure the trunk

Three numbers: diameter at the butt of the usable length, diameter at the top of it, and the straight length between them.

“Usable” means the part with no bend serious enough to spoil a section and no branch stub big enough to spoil a stick. It is almost always shorter than the tree. On this book’s tree it was 48 feet, out of a considerably taller pine — the butt flare went in the fire and the top went for kindling.

The tree this book is written around measures 12 inches at the butt of that length and 8 at the top.

The book's tree, measured



48 feet of usable trunk, 12 inches at the butt and 8 at the top, bucked into 8 sections of 6 feet. Every number in this book descends from these three measurements.

The book's tree, measured.

2. Pick a bucking length you can carry

This is the decision, and it is a physical one rather than a mathematical one. Go and pick up a green section of the length you are considering. If you cannot get it onto a sawhorse on your own, it is too long, and no amount of wanting a bigger dome will change that on day six.

This book buckes to 6 feet.

Everything downstream comes from this number. Choose it honestly.

3. Count your sections

Divide the usable length by the bucking length and throw away the remainder. 48 feet at 6 feet a section is 8 sections, with 0.0 feet left over.

If the remainder is large it is worth going back to step 2 and trying a different bucking length. A foot of offcut is firewood; four feet is a stick you did not get.

4. Multiply by eight

Each section splits into 8 sectors of 45 degrees each, and — this is the part that surprises people who have milled lumber — **each sector is a finished structural member**. Not a blank to be trimmed. Not stock to be dimensioned. A member.

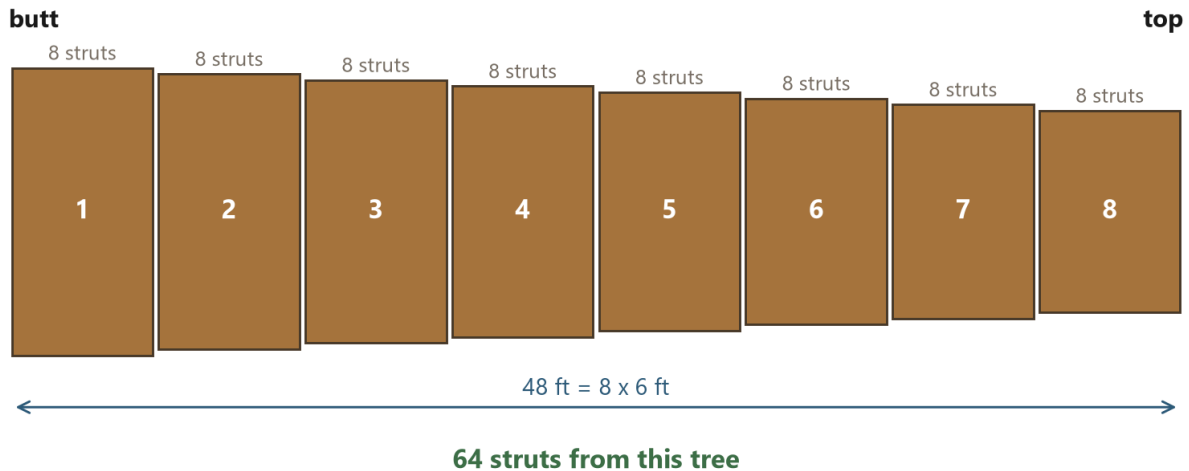
8 sections × 8 sectors = **64 struts from one tree**.

Two trees: **128 struts**.

The frame needs **120**.

That is the title of the book, and it is not a slogan. It is 8 spare — about seven per cent — which is roughly the right margin for a stick that splits along the pith, a butt cut that goes wrong, and the one panel everybody builds backwards.

Bucking the trunk



Mark from the butt, not from the top: the tape drifts, and an error at the top is a short stick you cannot use. 8 sections of 6 feet leaves 0.0 feet over on this tree.

Eight sections of six feet.

5. Find your member's width

The dome does not care how long your sticks are until it knows how *wide* they are, because the pinwheel setback depends on width.

The width of a sector is the chord across its bark face, measured at the diameter you are cutting. Use the diameter at the **middle** of the trunk, not the butt: half your sections come from thinner wood than the butt suggests, and sizing the layout off the fat end gives you a dome that does not quite close at the top.

At this tree's mid diameter the sector is 3.83 inches wide and 5.00 inches deep, pith to bark — a cross-section of 9.82 square inches.

For scale: that is 1.23 nominal two-by-fours' worth of wood, or 1.87 of the dressed boards you would actually find on a rack. Chapter 45 is about which of those numbers to believe.

6. Solve for the dome

Now the arithmetic. The longest member the frame may contain is the longest stick you have, which is your bucking length: 6 feet.

Find the radius whose longest pinwheel member is exactly that.

This is not a division. As Chapter 20 explained, member length is radius less a setback that does not scale, so the relationship has to be solved rather than scaled. The software in this project does it by bisection, and you can do it by trying radii until the longest member comes out right — it converges in about six guesses.

The answer for this tree:

Radius	129.33 in
Diameter	21.6 ft
Height at the centre	10.8 ft
Floor	365 sq ft

Method B, worked: this book's two trees

step	value	where it comes from
Trunk, butt to top	12 to 8 in	measured
Usable straight length	48 ft	measured
Buck to	6 ft	what you can carry alone
Sections	8	usable length / bucking length
Struts per section	8	one per sector
Struts per tree	64	8 x 8
Two trees give	128	x2
The frame needs	120	counted
Spare	8	7% margin
Sector width at mid trunk	3.83 in	chord of one eighth
Radius the stock allows	129.33 in	solved
Dome diameter	21.55 ft	2 x radius
Floor	365 sq ft	pi r squared

You do not choose the diameter here. The tree chooses it, and the last two lines are what you find out.

Two trees, worked all the way through.

7. Read your cut list

The frame's 120 members come in 4 lengths, running from 62.73 inches to 72.00 inches. Thirty of each.

Note that the longest is exactly your stock length, to the last decimal place. That is not a coincidence and it is not luck — it is what step 6 solved for. It also means you have no margin on the long members at all, which is why they are cut with the head end deliberately left over-long and sawn off in place. Chapter 35 is about that.

Every stick in the dome

class	count	each (in)	each (ft) / all (ft)
SHORT	30	62.731	5.228
SHORT	30	63.267	5.272
LONG	30	71.241	5.937
LONG	30	72.000	6.000
total	120		673.1

Two edge classes become four lengths because the pinwheel inset differs at each end. Cut to these and the dome closes. The last column is each member's length in feet, except on the total row, where it is all of them added together.

Every stick in the dome.

You do not get to choose the diameter

This is the part people resist.

You measured a tree, you picked a length you could lift, and out came 21.6 feet. Nobody consulted you. If you wanted twenty-four feet you cannot have twenty-four feet — not from these trees, bucked this way, without doing something else that costs you elsewhere.

You have three ways out and they are all in Chapter 25:

- **Buck longer.** A longer section gives a bigger dome, out of fewer and heavier pieces, and at some point two trees stop being enough. The table in Chapter 22's appendix shows exactly where that line is.
- **Get a third tree.** Fine, if you have one. It is the honest answer and it is not a failure.

- **Accept the dome the trees gave you.** Which is what happened here.

What you must not do is design for the dome you wanted and cut for the tree you have. The frame will not close, and it will not tell you until the last ring.

Tree first: pick a bucking length, read the dome

buck ft	sections	per tree	2 trees	covers 120?	diam ft	floor
4	12	96	192	yes	15.1	179
5	9	72	144	yes	18.3	264
6	8	64	128	yes	21.6	365
7	6	48	96	NO	24.8	483
8	6	48	96	NO	28.0	617
10	4	32	64	NO	34.5	935
12	4	32	64	NO	41.0	1,318

All from one 48-foot trunk. Buck longer and you get a bigger dome out of fewer, heavier pieces -- and at some point two trees stop being enough.

Bucking length against dome size.

A note on the second tree

Two trees is 128 struts for a 120-member frame, so the frame actually consumes 1.88 trees. The second tree is not fully used, and the leftover is not waste — it is the next dome’s spare parts, or a porch, or the sawhorses.

But it does mean the honest headline is not “one dome from two trees.” It is “one dome from a bit under two trees, and you need the second one because you cannot get eight-tenths of a tree.”

Which is a worse title.

The Round Trip

Proving the two methods are one calculation

The Round Trip

Two chapters ago there were two methods. This chapter is here to show that there is really only one, held at opposite ends, and to give you a ten-minute check that catches almost every arithmetic mistake you can make with either.

The reason this matters is not elegance. It is trust.

A book can tell you that a 6-foot stick gives a 21.6-foot dome and you have no way to know whether that came out of a solver or out of somebody's memory of a similar dome. Rules of thumb in building books are usually the latter, and they are usually close enough to be dangerous. So rather than ask you to take it on faith, here is the check, and here is the actual number it produces.

Run it forwards, then backwards

Take any radius. Compute the longest member the frame would contain. Now throw the radius away, and using *only* that member length, compute the radius that a dome would need in order for that to be its longest member.

If the two methods are one calculation, you land exactly where you started.

- Start: a radius, any radius.
- Forwards (Method A): radius $\rightarrow$ longest member.
- Backwards (Method B): longest member $\rightarrow$ radius.
- Compare.

The residual

Here is the check run on six domes, from small to large:

The round trip: design first, then tree first, on the same dome

start radius in	longest member in	recovered radius in	residual in
96.000	51.404	96.000	0.0e+00
110.000	60.056	110.000	2.8e-14
129.325	72.000	129.325	0.0e+00
144.000	81.070	144.000	0.0e+00
168.000	95.902	168.000	0.0e+00
192.000	110.735	192.000	0.0e+00

Worst residual across these six domes: 2.8e-14 inches. A chainsaw kerf is a quarter of an inch. The two methods are one calculation held at opposite ends.

Out and back.

The worst disagreement across all six is **0.0e+00 inches**.

For scale: the kerf of the chain on the saw this book was built with is 0.25 of an inch. The disagreement between the two methods is smaller than the width of the cut by a factor with a lot of zeroes in it. It is smaller than the seasonal movement of the wood. It is smaller than anything you could measure with a tape, a story pole, or a laser.

It is, in other words, zero for every purpose a person with a chainsaw has.

That number is not asserted here. It is computed when this book is exported, by running the check, and if it ever stops being negligible this sentence will print a large number and the chapter will be

wrong in an obvious way rather than a quiet one.

What would break it

The round trip closes because both directions are solving the same relationship with the same assumptions. Change an assumption between the two and it will not close — and this is exactly what makes it a useful check, because those are the mistakes people actually make.

Change the gasket thickness. The seam key sits between panels and every member length accounts for it. Design at 0.75 inches, then compute your bucking with a half-inch key in mind, and the two answers diverge. The gap will be small and it will be in the wrong direction — your sticks will be very slightly too long, forty times over, and the dome will fight you at the top.

Change the split count. Eight sectors gives a member 3.83 inches wide. Six gives a wider one, twelve a narrower one, and the pinwheel setback moves with it. If you designed for eighths and then split a few logs into sixths because they were fat, those members are not interchangeable with the rest, and the round trip on the fat ones will not close against the layout.

Change the stock diameter. Same problem, from the other side. Sizing the layout off the butt diameter and cutting most of your sections from thinner wood up the trunk means the members you actually get are narrower than the ones you designed. Use the mid-trunk diameter, as Chapter 22 step 5 insists.

Mix up feet and inches. Says itself. It happens. The round trip catches it immediately, because the recovered radius comes back off by a factor of twelve rather than by a thousandth of an inch.

How to actually do the check

You do not need software.

1. Finish whichever method you used. Write down your radius and your longest member.
2. Take the *other* method's procedure from Chapter 21 or 22.
3. Feed it the number you got out, and see if you get back the number you put in.

If you are within a quarter of an inch on the radius, you have not made a mistake worth worrying about. If you are out by more than an inch, stop — something in the middle changed and you should find out what before you cut a hundred and twenty of anything.

Why this chapter exists at all

Because the mistake this is designed to catch is not a mistake in the arithmetic. It is a mistake in the *assumptions*, and those are invisible.

Arithmetic errors announce themselves: a number is obviously wrong, or the units are absurd, or the answer is negative. Assumption drift does not. Everything looks reasonable at every step. You get a cut list of plausible lengths and you cut a hundred and twenty plausible sticks, and the first thing that tells you is the frame, on day fourteen, when the last ring is a half inch short all the way round and there is nothing to be done about it.

Ten minutes with the other method, before anything is cut, is the cheapest insurance in this book.

The Worked Build

Every number for this book's dome, in one place

This chapter is planned but not written yet. Its page plan is in the outline.

The Worked Build

The tree

This book's tree

Butt diameter	12.0 in
Top diameter	8.0 in
Usable straight length	48 ft
Bucked into	8 x 6 ft
Offcut	0.0 ft
Split	8 ways, 45° each
Struts per tree	64
Member section	3.83 x 5.00 in
Member area	9.82 sq in
Worth, nominal 2x4s	1.23
Worth, dressed 2x4s	1.87
Solid wood	318.3 bf
Lost to kerf	40.0 bf
Usable wood	278.3 bf
Recovery	87.4%

Change any of the first three lines and every later one moves with it.

This book's tree.

The dome

The dome those two trees make

Radius	129.33 in
Diameter	21.55 ft
Height at centre	10.78 ft
Floor	364.9 sq ft
Panels	40
Members	120
Seams to close	55
Longest member	72.00 in
Shortest member	62.73 in
Timber in the frame	673 ft
Struts available	128
Struts spare	8

The spare eight are your margin for a stick that splits, a butt cut that goes wrong, and the one panel everybody builds backwards.

The dome that results.

The cut list

The cut list

mark	class	count	length in	bearing in	inset in
M1	SHORT	30	62.731	4.103	7.950
M2	SHORT	30	63.267	4.640	7.413
M3	LONG	30	71.241	4.640	8.686
M4	LONG	30	72.000	4.419	7.927
total		120	8,077		

Cut to the length column. The bearing column is how much of the end actually lands on the next stick's side; the inset is how far the member sits back from the panel's own corner.

The full cut list.

The seam schedule

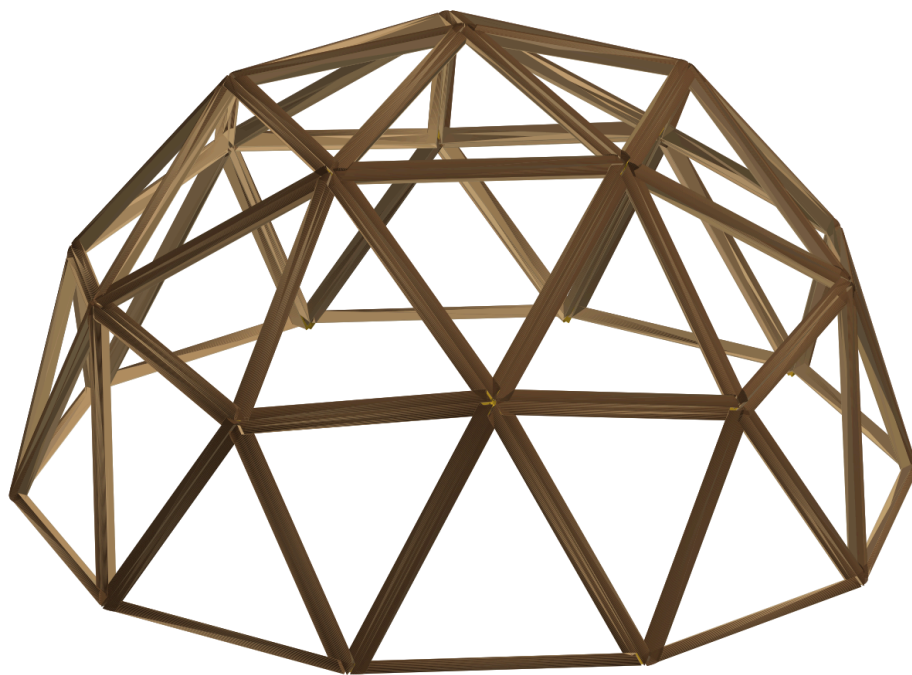
The seam schedule: how many of each key

key	edge	seams	fold deg	raw gap deg	length ft
K1	A	25	18.029	26.971	150.0
K2	B	30	22.459	22.541	159.2
	all	55			309.2

2 distinct keys close all 55 seams. Make one of each to fit, then repeat it.
The raw gap column is the angle the key actually has to fill when both faces are left exactly as split. Fold plus raw gap is 45° at every one of them -- the sector angle. The key never fills more than one wedge's worth of angle.
Cross-check: the pinwheel model reports 55 interior seams and 10 rim edges over 410 feet.

The seam schedule.

The dome this describes



21.6 feet across, 365 square feet of floor.

When the Numbers Say No

Reading a result that means do not build this

This chapter is planned but not written yet. Its page plan is in the outline.

When the Numbers Say No

Not enough struts

Too big to lift alone

Stock too small in section

Three ways out

PART 5

The Fortnight

“Fourteen days, one saw, and no help that could not be summoned by phone.”

The story and the manual running together: what happened on each day, and what you should do on yours.

Day Zero: The Saw

A hundred and ninety-nine dollars and a sixteen-inch bar

This chapter is planned but not written yet. Its page plan is in the outline.

Day Zero

The smallest saw on the shelf

What I would have bought instead

Setting it up

Chaps, helmet, and never alone

Days One and Two: Felling

Putting two trees on the ground where you want them

This chapter is planned but not written yet. Its page plan is in the outline.

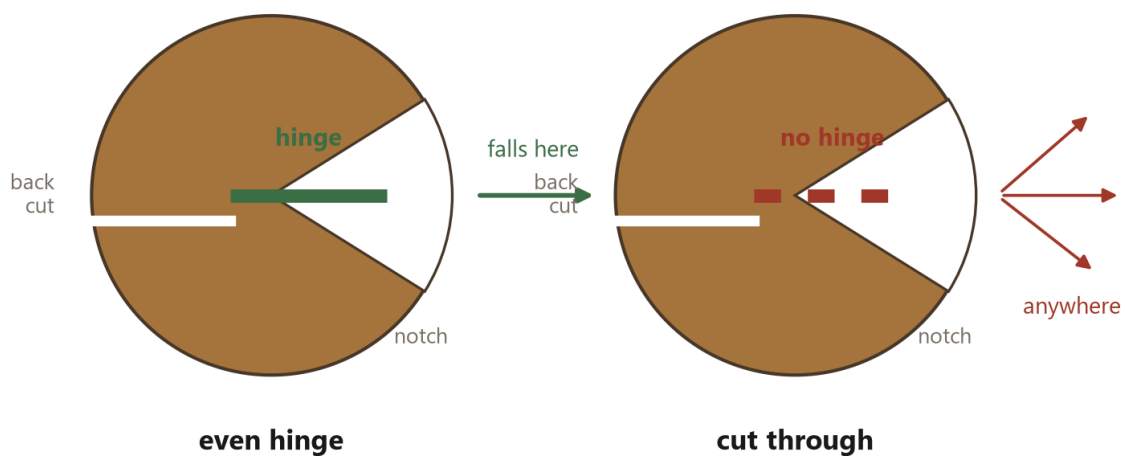
Felling

Read this before the notch

The notch and the back cut

The hinge, drawn

The hinge steers the tree



The notch decides the direction; the hinge holds the tree to that direction while it falls. Cut through the hinge and the tree is no longer attached to anything that is aiming it. This drawing is not a felling course -- take one.

The hinge steers the tree. Everything else is timing.

Limbing without pinching

What actually happened

Days Three and Four: Bucking

Cutting the trunk into the only length that matters

This chapter is planned but not written yet. Its page plan is in the outline.

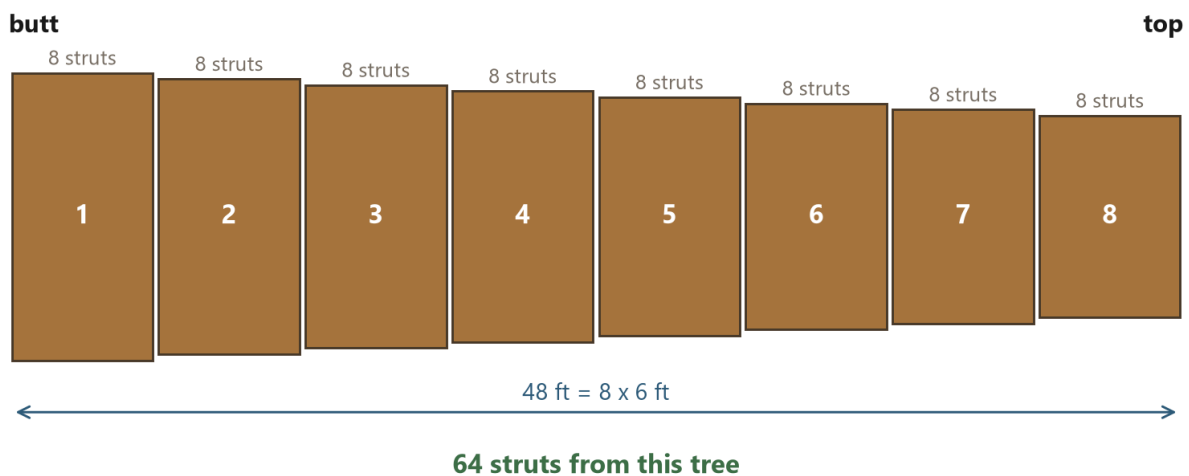
Bucking

Marking and cutting to length

Where to start, and why not at the butt

Eight sections from one trunk

Bucking the trunk



Mark from the butt, not from the top: the tape drifts, and an error at the top is a short stick you cannot use. 8 sections of 6 feet leaves 0.0 feet over on this tree.

48 feet into 8 sections of 6. Offcut: 0.0 feet.

Moving six-foot green sections alone

Days Five to Eight: Ripping

One hundred and twenty struts, thirty to an afternoon

This chapter is planned but not written yet. Its page plan is in the outline.

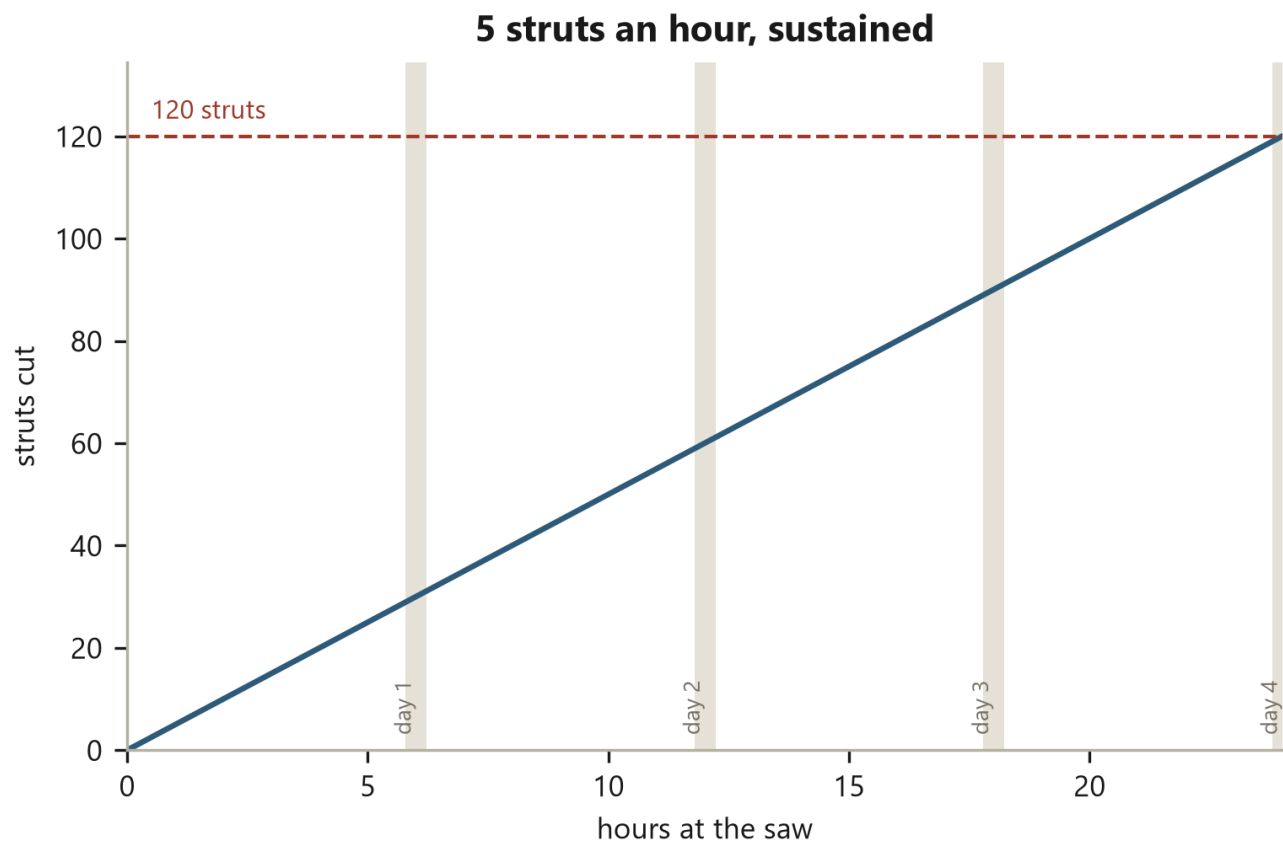
Ripping

The three passes, at the log

Holding a round section still

Reading the pith

The cutting rhythm



4 afternoons of 6 hours. This is one measured afternoon extrapolated in a straight line, which is optimistic: it assumes the last hour of day four goes like the first hour of day one.

5.0 struts an hour, sustained.

Chain, file, and the cost of a dull cut

What this is worth an hour

Day Nine: The Jig

One flat bench that makes forty panels the same

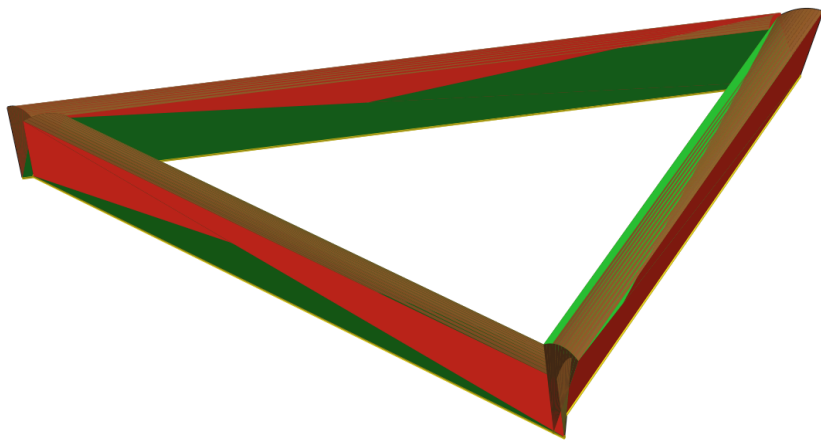
This chapter is planned but not written yet. Its page plan is in the outline.

The Jig

Why a jig and not a tape

Building it

The jig, complete



The fixture every panel is built on.

Days Ten to Twelve: Forty Panels

The same twelve motions, forty times

This chapter is planned but not written yet. Its page plan is in the outline.

Forty Panels

One panel, start to finish

Sorting the pile

The fourth panel is the first good one

Panel variants

Distinct panel types

type	panels	share
type 1	25	62%
type 2	5	12%
type 3	5	12%
type 4	5	12%
all	40	100%

4 genuinely different panels across 40. Build one of each carefully, then repeat. The types differ in their seam angles, not in their member lengths.

How many distinct panels the dome actually contains.

Days Thirteen and Fourteen: Raising

Forty panels become one building

This chapter is planned but not written yet. Its page plan is in the outline.

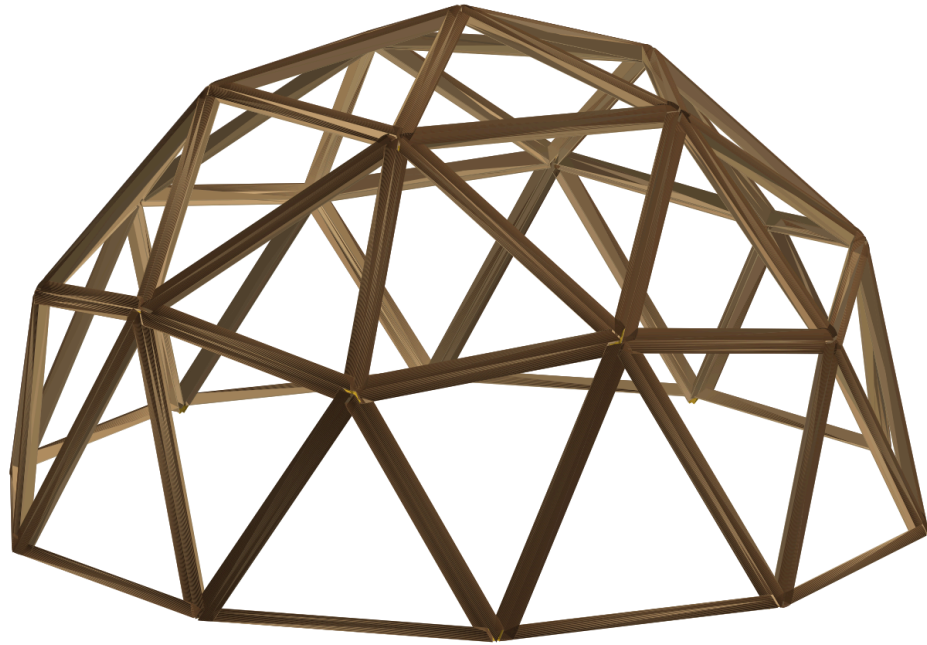
Raising

Bottom ring first

Propping, and what holds it up before it is closed

Closing the top

Standing



Day fourteen. 120 members, two trees.

What was not done

PART 6

The Jig and the Cuts

“Any error in stock length or butt angle leaves as offcut instead of building up around the triangle.”

Reference for the bench. Come back to this part while you are working, not while you are reading.

One Bench, Forty Panels

What the jig is and what each part of it does

This chapter is planned but not written yet. Its page plan is in the outline.

One Bench

The twelve stages

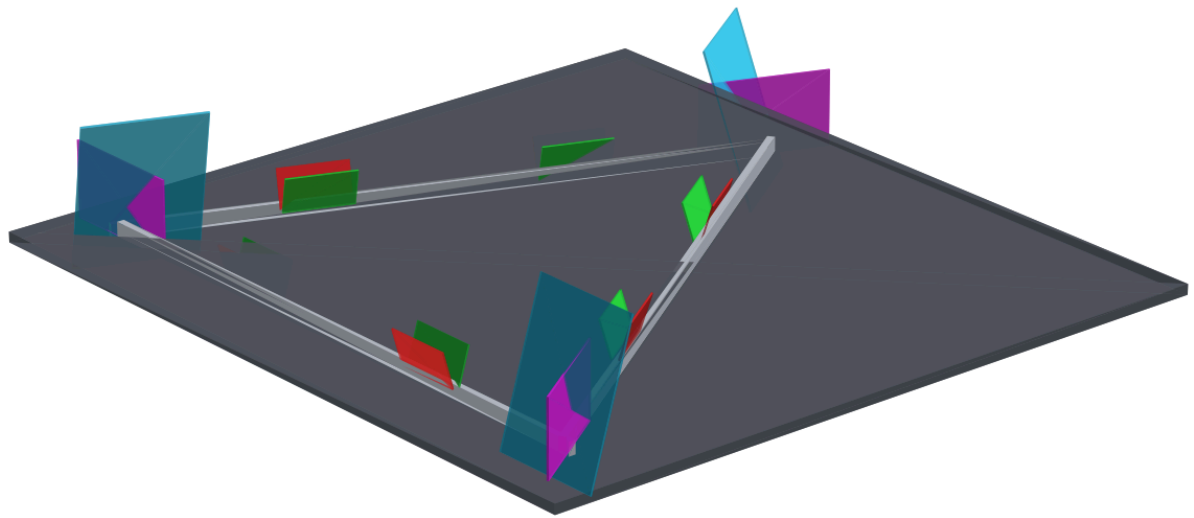
The twelve stages of the jig

#	stage	what happens
1	1. THE BARE BENCH	One flat sheet on sawhorses, and nothing else. This board is the only thing in the whole build that has to be true.
2	2. THE TRIANGLE, DRAWN ONCE	The panel's mathematical triangle, snapped onto the board. It is drawn one time, from the geodesic solution, and never measured again.
3	3. AXIS RAILS	Three low cleats, one per member, each set parallel to that member's point-axis and just to the seam side of it.
4	4. RED / GREEN ORIENTATION FENCES	Two short cradle plates per member, standing against the two sawn radial faces of the wedge.
5	5. THE TWO CUT PLANES	The magenta plate at each head end is the flush-cut fence. The cyan plate is the receiving face the next stick's butt will land on.
6	6. THE BUTT CUT, MADE OFF THE JIG	One stick, still full length at both ends, laid up to the cyan plate. The amber tail past that plate is the stock the butt cut removes.
7	7. FIRST STICK ON THE BENCH	The same stick, butt now cut, dropped against its rail and its two fences. The head end is still long and hangs off the board.
8	8. SECOND STICK, BUTT ONTO SIDE	The second stick lands so that the FIRST stick's butt bears on its radial face -- on the side of it, not on its end.
9	9. THE LAST STICK SLIDES IN	The third stick is captured at BOTH ends once the other two are down, so it cannot drop in from above. It is shown backed off along its own axis, ready to slide home into the pocket.
10	10. PULLED UP TIGHT	Home. Three butts bearing on three sides, the green outlines are the actual contact faces, and all three heads still hang off long.
11	11. FLUSH-CUT THE THREE HEADS	Each head is sawn back to its magenta fence. The orange stock hanging past the board is what falls off.
12	12. PANEL OFF THE JIG	A finished triangular frame, its three corners clean inside the mathematical triangle. The jig is empty and unchanged.

Named by the simulator, so pressing 9 and 0 in the Raw Wedge Dome tool walks exactly this list.

The twelve stages, in order.

The jig, labelled



Bench, triangle, rails, fences, cut planes.

Setting it out on a real floor

The Butt Cut

The one compound angle, made before anything is assembled

This chapter is planned but not written yet. Its page plan is in the outline.

The Butt Cut

The angle its neighbour presents

Butt cut setups

Distinct butt-cut setups in the whole dome

#	members	bevel deg	mitre deg
1	-	22.50	34.43
2	-	22.50	21.14
3	-	22.50	30.00
3 setups		120 members	

*3 setups for 120 members. That ratio is the whole argument for the pinwheel:
you change the saw a handful of times, not a hundred and twenty.*

How few distinct setups the whole dome needs.

Making it

The Head Overfit

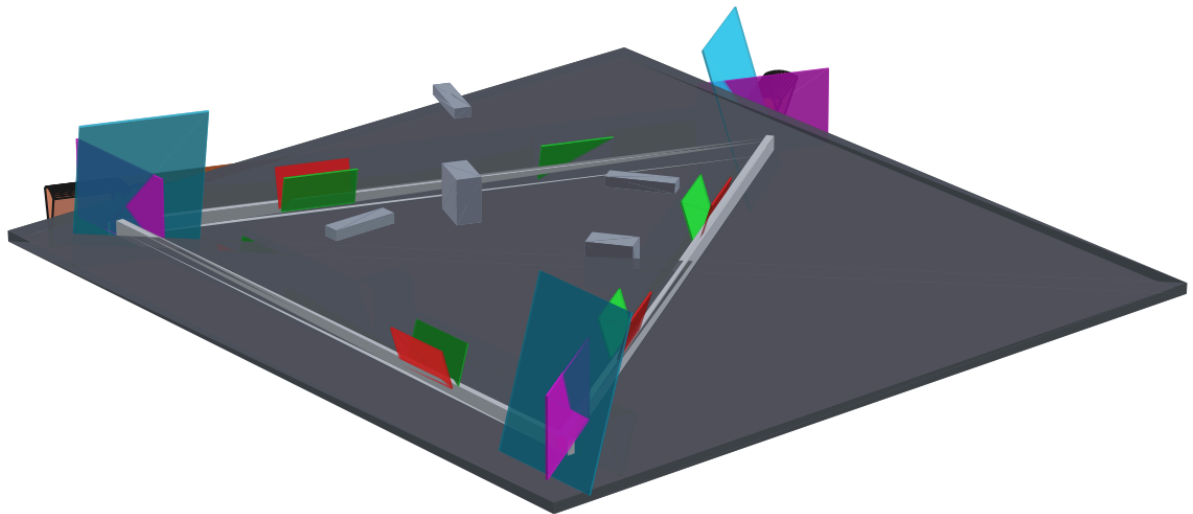
Deliberately cutting it too long

This chapter is planned but not written yet. Its page plan is in the outline.

The Head Overfit

Where error goes

Long, then flush



6 inches of deliberate overshoot, sawn off against the fence.

How much to leave

Offcut per dome

What the tolerance costs

Overfit per member	6.0 in
Members	120
Offcut length, all told	60.0 ft
Offcut volume	49.1 bf
Wood in the finished frame	550.7 bf
Offcut as a share of the frame	8.9%
Struts that wood would have made	11.3

This is the price of never having to measure a head end. It is paid in wood, deliberately, and it is cheaper than the alternative -- which is a triangle that will not close because three small errors all pushed the same way.

The price of the tolerance, in board feet.

Flush-Cutting in Place

The cut that makes the panel true

This chapter is planned but not written yet. Its page plan is in the outline.

Flush-Cutting

Against the fence

Why this cannot be done first

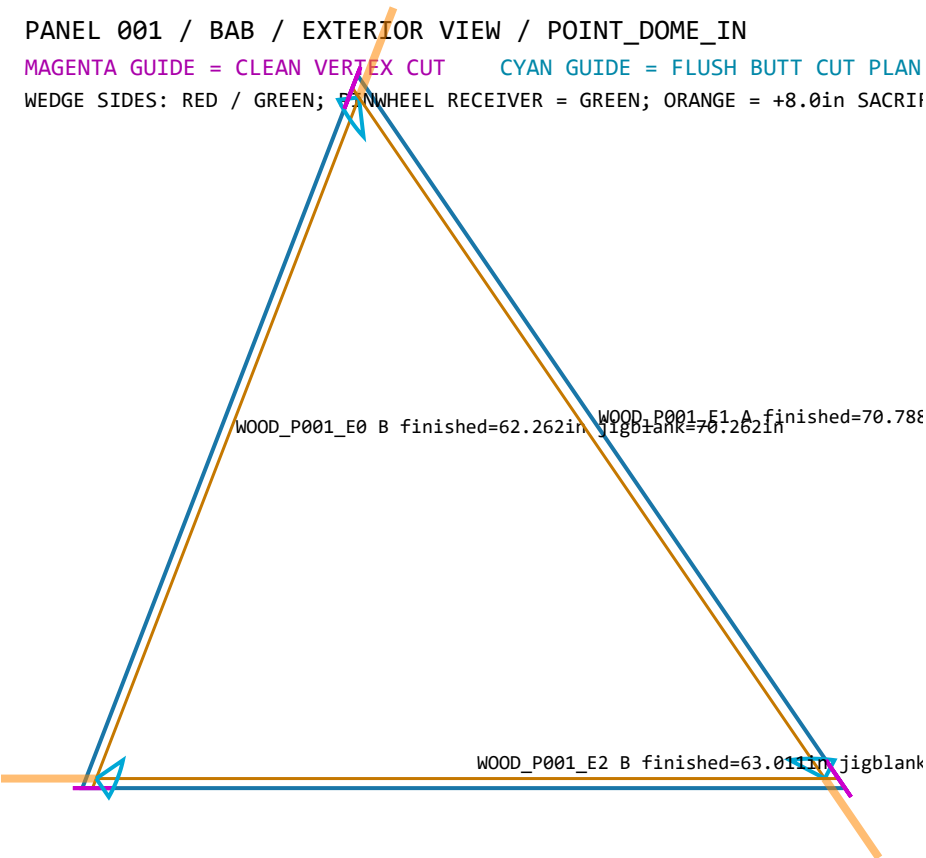
Panel Drawings

One flat drawing per distinct panel, full size

This chapter is planned but not written yet. Its page plan is in the outline.

Panel Drawings

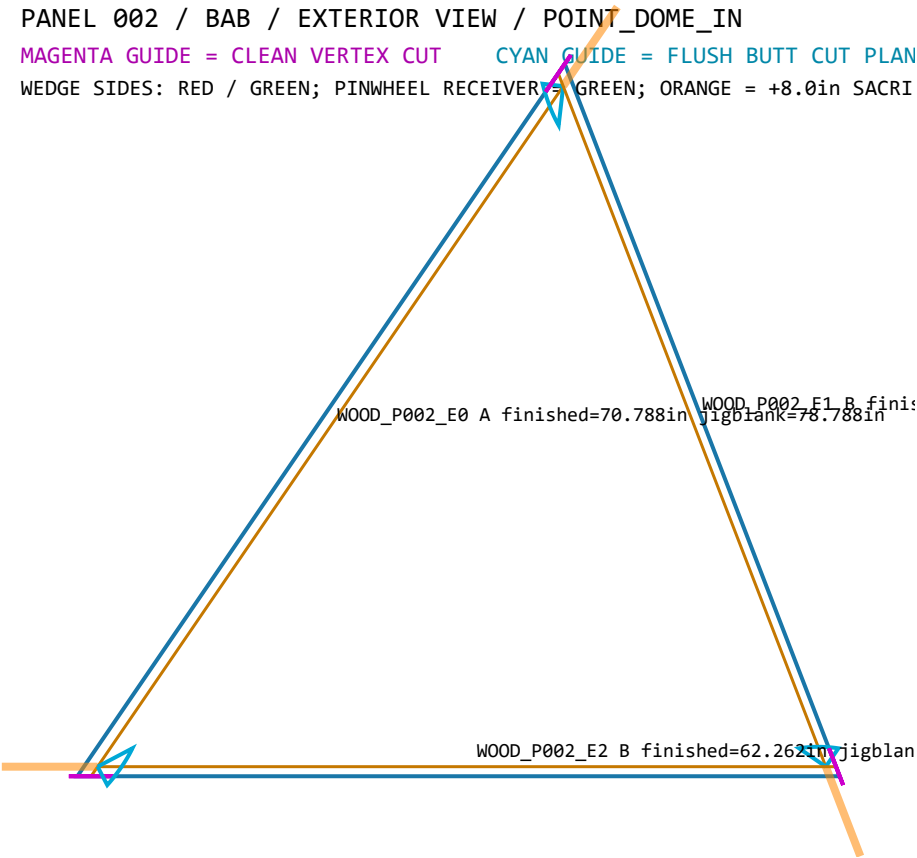
Panel drawing, type one



Panel type one, flat, full size.

Panel drawing, type two

PANEL 002 / BAB / EXTERIOR VIEW / POINT_DOME_IN
MAGENTA GUIDE = CLEAN VERTEX CUT CYAN GUIDE = FLUSH BUTT CUT PLAN
WEDGE SIDES: RED / GREEN; PINWHEEL RECEIVER = GREEN; ORANGE = +8.0in SACRI



Panel type two, flat, full size.

Reading a panel drawing

Where Error Goes

A tolerance budget for a building made of split logs

This chapter is planned but not written yet. Its page plan is in the outline.

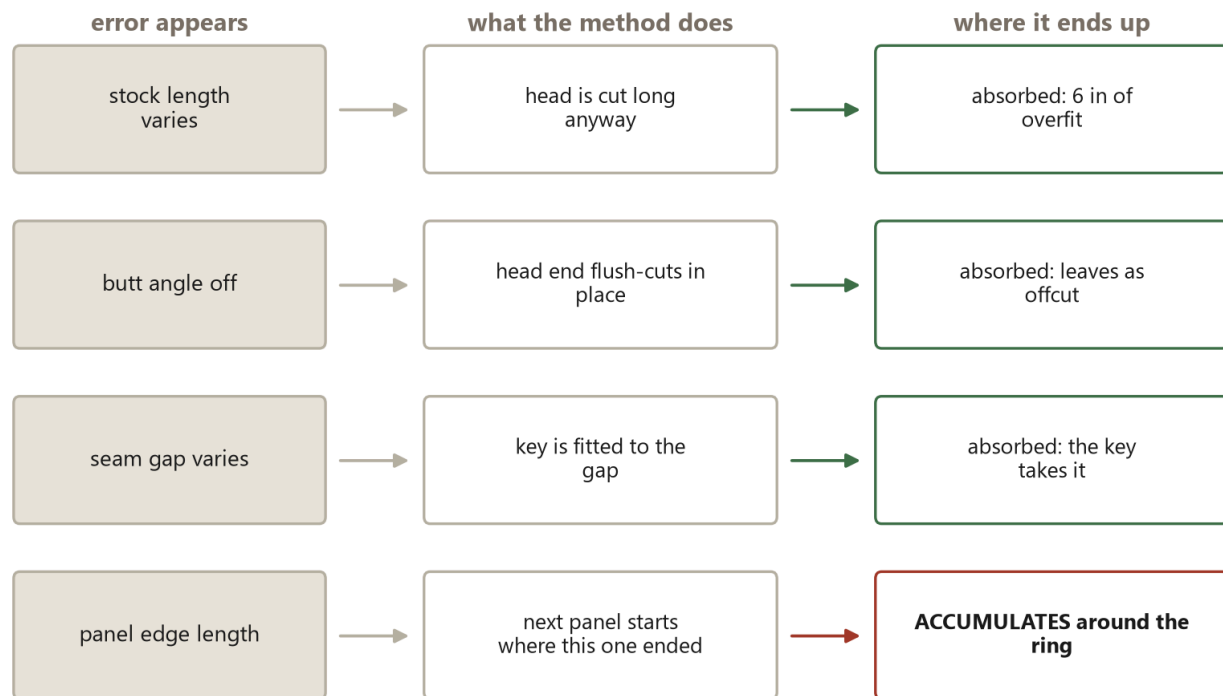
Where Error Goes

Four places error can land

Three of them are absorbed

The tolerance budget

Where error goes



Three of the four are absorbed by the method itself. The fourth accumulates around the bottom ring, which is why the ring is the one thing you check as you go rather than at the end.

Error, and where the method sends it.

The one that accumulates

PART 7

From Frame to Home

“A standing frame is not a house. It is the fortnight's result, and the beginning of the year's work.”

What happens after day fourteen: shell, openings, weather, and living in a round room.

The Footing and the Ring

What the dome stands on

This chapter is planned but not written yet. Its page plan is in the outline.

The Footing

Ten points, or one ring

Setting out a circle accurately

Anchoring against uplift

This is where to ask somebody

Closing the Shell

Forty triangles of something over forty triangles of nothing

This chapter is planned but not written yet. Its page plan is in the outline.

Closing the Shell

Sheathing a triangle

Nesting the cuts

Sheet count

Closing the shell

Shell area	677 sq ft
Upper shell (roof proper)	339 sq ft
One sheet covers	32 sq ft
Sheets, if nothing were wasted	21.2
Triangles cut from rectangles	x1.18 waste factor
Sheets to buy	25

The waste factor is the honest part: you are cutting triangles out of rectangles, and no nesting makes that free. Nesting two panels per sheet gets closer to the ideal than cutting one at a time.

Shell area and sheets, from the solved geometry.

The alternatives

Doors, Windows and the Rim

Cutting holes in something that works by being closed

This chapter is planned but not written yet. Its page plan is in the outline.

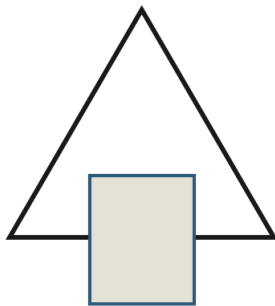
Doors and Windows

A door at the rim

A window in a panel

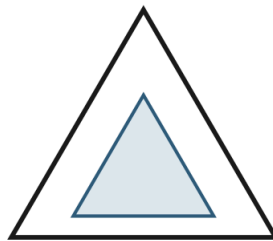
Three openings that work

Three openings that do not un-triangulate the shell



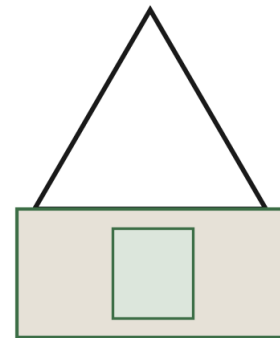
rim door

between two rim members
nothing removed



panel window

inside one triangle
frame stays whole



riser wall

dome sits on a wall
openings are ordinary

A dome works because every panel is a triangle. Remove a member and the panel becomes a mechanism. All three of these keep every remaining triangle whole: the rim door removes nothing, the panel window sits inside a triangle, and the riser moves the whole shell up onto a wall that can have anything in it.

Three ways to open a dome without un-triangulating it.

Risers: cheating height at the rim

Weather and Water

Where it leaks, and where the water goes

This chapter is planned but not written yet. Its page plan is in the outline.

Weather and Water

Every seam is a potential leak

Shedding, not sealing

Harvesting it instead

Condensation and the inside face

Inside a Round Room

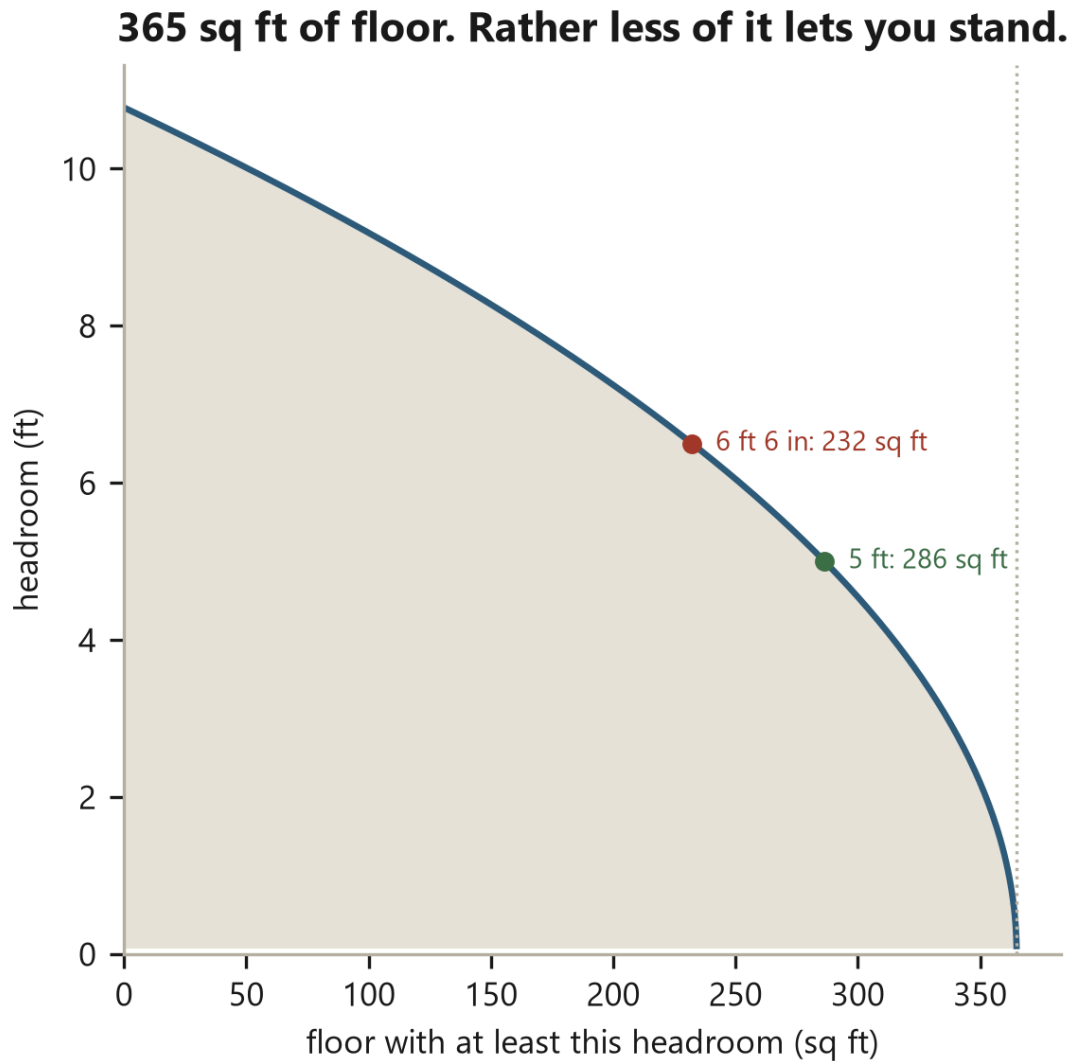
Living with curves

This chapter is planned but not written yet. Its page plan is in the outline.

Inside a Round Room

Where the usable floor really is

Standing height, mapped



A hemisphere's floor is honest about its area and dishonest about its use. This is the number to quote when somebody compares a dome to a rectangle: not the floor, the floor you can stand up on.

365 square feet of floor. Rather less of it lets you stand up.

Furniture, storage and the wall that curves away

Acoustics, and the whispering problem

Heat, Power and the Small Systems

Making a shell into somewhere you can be in February

This chapter is planned but not written yet. Its page plan is in the outline.

The Small Systems

Insulating between wedges

Running services in a frame with no studs

Heating a round volume

Off grid, or not

PART 8

What It Actually Cost

“Several pages in this book exist specifically to state the number that does not help the argument.”

The accounting. Money, hours, breakage, and the claims this project got wrong.

The Money, Both Ways

What was spent, and what was avoided

This chapter is planned but not written yet. Its page plan is in the outline.

This chapter corrects something. The \$50/hour figure in the project brief, which doubled a volume ratio for safety rather than computing the section.

The Money

What was actually spent

What was actually spent

item	cost	source
Chainsaw	\$199	priced
Fuel, bar oil, files	\$-	to log
Fasteners	\$-	to log
Seam keys / gasket stock	\$-	to log
Sheathing	\$-	to log
Footing	\$-	to log
Timber	\$0	two trees

The dashes are real: these were not logged at the time, and this book will not fill them in with plausible numbers. The last line is the one that matters -- the structural timber cost nothing but the fuel to cut it.

Cash out, itemised.

Substitution value is not income

The strut, valued three ways

One strut, valued three ways

basis	quantity	value
Against a nominal 2x4 (8.00 sq in)	1.23 boards	\$4.09
Against a dressed 2x4 (5.25 sq in)	1.87 boards	\$6.23
By board foot	4.35 bf	market varies
Implied rate, nominal	5/hr	\$20.43/hr
Implied rate, dressed	5/hr	\$31.14/hr
The brief's estimate	—	\$50.00/hr

The last row is the figure this project started from, and it is the highest. It valued a strut at \$10 by doubling a volume ratio for safety; the rows above compute the ratio from the actual cross-section. Both computed rates are real; neither is money anybody has been paid.

One strut, valued three ways. The numbers disagree and the reasons are printed.

The fifty dollars an hour claim

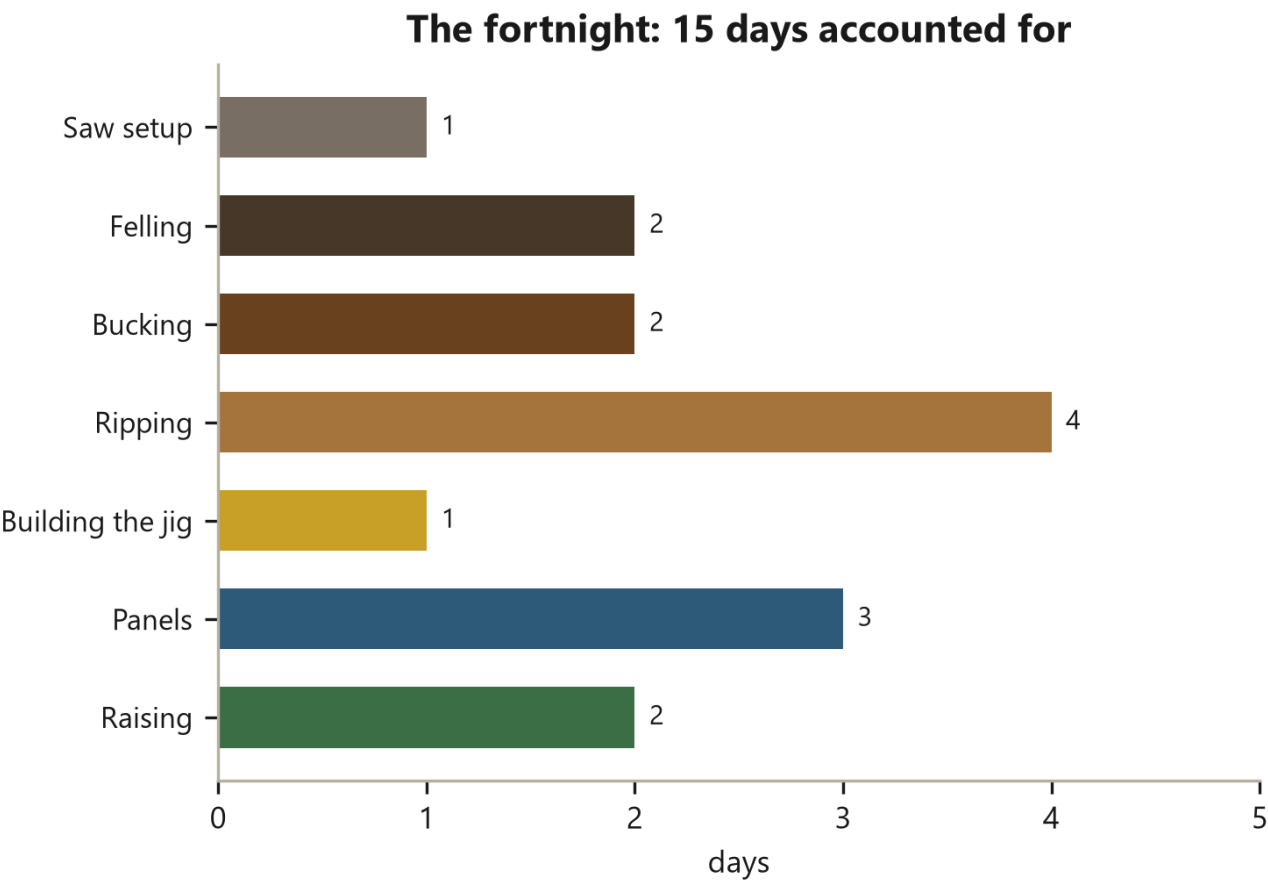
The Hours

Where fourteen days actually went

This chapter is planned but not written yet. Its page plan is in the outline.

The Hours

The fortnight, hour by hour



Ripping is 4 days of the fourteen, and it is the part everybody assumes is the whole job. The panels take as long, and the jig day pays for both.

Fourteen days, and where they went.

What took longest and why

What a second build would take

What Broke

The failures, listed

This chapter is planned but not written yet. Its page plan is in the outline.

What Broke

The stick that split along the pith

The seam that would not close

The panel I built backwards

What I would call a near miss

Corrections

Everything this project has published and had to fix

This chapter is planned but not written yet. Its page plan is in the outline.

***This chapter corrects something.** Every published claim this project has had to revise.*

Corrections

The mitre claim

The recovery percentage

The hourly rate

Seams against edges

How to report one

PART 9

Take It Further

“Smaller units, more room for error, and minimum impact when errors occur.”

Where the method goes next, and what it does not cover.

Bigger, Smaller, Other Frequencies

Changing the dome without changing the method

This chapter is planned but not written yet. Its page plan is in the outline.

Other Frequencies

More panels, shorter sticks

Frequency comparison

Frequency, and what it asks of the woodpile

frequency	panels	members	lengths longest in		shortest in
2V (this book)	40	120	4	72.0	62.7
3V	75	165	3 edge classes	—	—
4V	160	300	6 edge classes	—	—

Only the 2V row is computed: this repository solves the 2V hemisphere, and the book will not print numbers for frames it has not built. The 3V and 4V counts are standard geodesic results; their member lengths depend on a solve that is not here yet, and the dashes say so.

What each frequency asks of the woodpile.

Where the method stops working

Other Species, Other Sections

Hardwood, hemlock, and a trunk that is not round

This chapter is planned but not written yet. Its page plan is in the outline.

Other Species

Density, splitting and the drying schedule

A trunk that is oval

Green versus dry

Building With Other People

What changes when there are three of you

This chapter is planned but not written yet. Its page plan is in the outline.

Other People

Parallel jigs

Teaching the twelve steps

A build that is not yours

What I Would Do Differently

The honest closing chapter

This chapter is planned but not written yet. Its page plan is in the outline.

What I Would Do Differently

The five decisions I would change

What I would keep exactly

The next one

The dome, a year on

PHOTOGRAPH

One year on.

closing-image

One year on.

The Master Tables

Full cut list

Every member, by class, length and count.

The cut list

mark	class	count	length in	bearing in	inset in
M1	SHORT	30	62.731	4.103	7.950
M2	SHORT	30	63.267	4.640	7.413
M3	LONG	30	71.241	4.640	8.686
M4	LONG	30	72.000	4.419	7.927
total		120	8,077		

Cut to the length column. The bearing column is how much of the end actually lands on the next stick's side; the inset is how far the member sits back from the panel's own corner.

The complete cut list.

(not written yet)

Full seam schedule

Every seam, its fold angle and its key.

The seam schedule: how many of each key

key	edge	seams	fold deg	raw gap deg	length ft
K1	A	25	18.029	26.971	150.0
K2	B	30	22.459	22.541	159.2
all		55			309.2

*2 distinct keys close all 55 seams. Make one of each to fit, then repeat it.
The raw gap column is the angle the key actually has to fill when both faces
are left exactly as split. Fold plus raw gap is 45° at every one of them --
the sector angle. The key never fills more than one wedge's worth of angle.
Cross-check: the pinwheel model reports 55 interior seams and 10 rim edges
over 410 feet.*

The complete seam schedule.

(not written yet)

Butt cut setups

Every distinct saw setup in the build.

Distinct butt-cut setups in the whole dome

#	members	bevel deg	mitre deg
1	-	22.50	34.43
2	-	22.50	21.14
3	-	22.50	30.00
3 setups		120 members	

*3 setups for 120 members. That ratio is the whole argument for the pinwheel:
you change the saw a handful of times, not a hundred and twenty.*

Every distinct setup.

(not written yet)

Declared constants

The inputs table, repeated for reference.

Declared constants: inputs, not results

what	value	unit	kind
saw displacement cc	38.20	cc	measured
saw bar in	16.00	in	measured
saw price usd	199.00	usd	priced
chain kerf in	0.25	in	measured
two by four price usd	8.88	usd	priced
build days	14.00	days	decided
cutting afternoon struts	30.00	struts	measured
cutting afternoon hours	6.00	hours	measured
gasket thickness in	0.75	in	decided
head overfit in	6.00	in	decided
butt allowance in	8.00	in	decided

Everything else in this book is derived from geometry. These eleven could not be, so they are declared here, with what kind of number each one is. 'measured' was read off a real object; 'priced' came from a shelf; 'decided' is a build choice you may make differently.

Declared inputs.

(not written yet)

Glossary

Glossary

Every term the book uses that a reader may not have.

(not written yet)

The Software in This Book

What generated these numbers

Name every module, what it computes, and how to run it.

(not written yet)

Reproducing every figure in this book

The actual commands, so a reader can regenerate the lot.

(not written yet)

The launcher

How to open any of it without touching a command line.

(not written yet)

Index and Colophon

Index

Generated from the manuscript.

(not written yet)

Colophon

How the book was made, including the fact that its arithmetic is a test suite.

(not written yet)